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PALAEOMAGNETISM OF THE SVECONORWEGIAN MOBILE BELT OF THE FENNOSCANDIAN SHIELD

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ABSTRACT

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The Sveconorwegian belt is a broad terrain in the south western sector of the Fennoscandian Shield subjected to thermal activation, widespread intrusion and limited tectonic activity between ca. 1100 and 950 Ma, and magnetised during slow uplift and cooling following these mobile events. This study comprises a palaeomagnetic investigation of four areas spanning the zone: the anorthosite, farsundite and related intrusive complexes of southwest Norway, the gabbro—norite intrusions of the Bamble sector bordering the Oslo Graben, the gabbro—anorthosite complex at Brattön and Älgön in west Sweden, and dolerite intrusions near the eastern tectonic front to the zone in Sweden. Site mean directions and demagnetisation trends in the Rogaland intrusive complex define an apparent polar wander (a.p.w.) swathe of just over 30° of arc; poles fall along this swathe according to their position in the complex and their blocking temperatures, and the resultant path is interpreted as a migration of this zone away from the palaeopole at a rate of 1.3—3.0° Ma $^{-1}$. The total swathe represents 10—20° of a.p.w. movement at 940—900 Ma; an isolated later ('Y') magnetisation may represent a later extension of the path at ca. 840 Ma. Two sets of magnetisations, a NW negative, predominantly hematite-held, and SSW negative, predominantly magnetite-held, are recognised in the gabbro—norite intrusions of the Bamble sector and are linked to radiometric events at 1090 and 1010 Ma. In the Brattön and Älgön norite—anorthosite bodies of west Sweden, demagnetisation trends and site mean directions define a short swathe of westerly negative directions linked to radiometric ages averaging ~ 1000 Ma. Minor dolerite (hyperite) bodies intruded near the eastern margin of the Sveconorwegian belt possess predominantly steep positive and negative directions of magnetisation comparable to results from the Scania region at the southern margin of the belt.

The collective palaeomagnetic data from the Sveconorwegian belt and its borders define a flattened anticlockwise a.p.w. loop with an amplitude of ~ 70° and executed over the approximate interval 1050—840 Ma with the apex, or hairpin, defined by the Rogaland data at ca. 940 Ma. This loop correlates precisely with a loop defined by the uplift and cooling magnetisations from the Grenville terrain of the Laurentian Shield and shows that the Fennoscandian and Laurentian Shields were in juxtaposition between 1050 and 850 Ma with the two linear frontal zones disposed at a high angle to one another. These fronts were probably initiated in alignment, however, because they are disposed in this way on a primitive reconstruction derived from the pre-1190 Ma palaeomagnetic data.

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Break up and relative movements of the two shields occupied a short period between 1190 and 1050 Ma, following the Grenville—Sveconorwegian tectonism. It was associated with widespread rifting, dolerite intrusion and alkaline magmatism into the already stabilised parts of the shields.

INTRODUCTION

Tectonic setting of the Sveconorwegian belt

It has been recognised for two decades that the Fennoscandian Shield subdivides into three major structural—geochronological divisions (Polkanov and Gerling, 1961; Kratz et al., 1968), although more refined and detailed analyses recognise the complexity of these zones and identify the defining thermal—tectonic events as only the last of several episodes responsible for growth and consolidation of each terrain. The oldest (Saamokarelian) terrain incorporates the north western part of the Shield stabilised prior to 2600 Ma, although some overprinting between 2500 and 1800 Ma is also evident. The Svecofennian terrain, incorporates most of the Precambrian basement in Sweden while the Svecokarelian terrain includes comparable zones in south east Finland profoundly affected by tectonic activity at ~ 1800 Ma and widely intruded by granitic magmas for 200—300 Ma after this. The youngest of the three divisions comprises the Sveconorwegian terrain of southwest Sweden and southern Norway, mobilised between 1100 and 950 Ma. This belt is cut off in the north by the Scandinavian sector of the Caledonian orogenic belt and in the east it is separated from the older Svecofennian terrain by the 20—30 km wide Schistosity Zone [otherwise known as the Protogine Zone or Sveconorwegian (Gothide) Front].

This zone of highly foliated rocks separates the Sveconorwegian block to the west, yielding a range of Rb—Sr ages between 1700 and 950 Ma (Neumann, 1960; Broch, 1964; O'Nions et al., 1969; O'Nions and Baadsgaard, 1971, O'Nions and Heier, 1972; Versteeve, 1975) and K—Ar minerals ages concentrating between 1000 and 800 Ma (Versteeve, 1975; Åberg, 1978), from the Svecofennian terrain in the east where only local resetting of mineral ages occurred after 1700—1500 Ma. The borders of the latter zone were, however, intruded by a suite of 950—850 Ma front-parallel dykes into tensional fractures that appear to be related to post-tectonic uplift of the Sveconorwegian terrain (Patchett and Bylund, 1977). Most recent geochronological assessments of data from the Sveconorwegian belt (e.g., Field et al., 1982) emphasise the importance of gneiss-forming events within a basement of Svecokarelian age: much of the Sveconorwegian episode appears to relate primarily to an uplift and cooling event, although there is no doubt that the western part in southwest Norway was profoundly mobilised at 1100—1000 Ma.

Previous palaeomagnetic studies

Since the mineral age record shows that the rocks within this belt were uplifted and cooled following the ca. 1100 Ma mobile episode, the magnetisations recorded by these rocks have been widely related to the interval of post-tectonic uplift and cooling, and they represent the major opportunity for reconstructing the apparent polar wander (a.p.w.) record for the Fennoscandian Shield during this time interval. In a wider context, this period is important because it is also represented by uplift of the Grenville Belt in the Laurentian Shield (Baer, 1976) and for some years tectonic resolution of palaeomagnetic data from this latter belt has been hampered by the paucity of contemporaneous rocks from elsewhere (McWilliams and Dunlop, 1978; Piper, 1980). As Patchett and Bylund (1977) have shown, this problem can be partly surmounted in the Fennoscandian Shield by the front-parallel dykes intruded into the previously-stabilised shield adjacent to the mobile belt.

Mulder (1971) reported palaeomagnetic results for 6 dolerite (hyperite) dykes intruded into the Schistosity Zone in southern Sweden; K—Ar ages of 1573—781 Ma were obtained with variable directions of magnetisation of both polarities. A notable aspect of this study was an analysis of magnetic directions across a 35 m wide dyke at Hägghult which identified a sequential decrease in magnetic inclination from $-78°$ at the margins to $-72°$ in the interior. Further results for hyperite bodies of the Swedish (2 sites) and southeast Norway (4 sites) sectors of the Sveconorwegian belt are reported by Poorter (1975). Abrahamsen (1974) studied related WNW-trending dykes near Gothenburg and made a detailed study of 4 dykes on the island of Bornholm (Abrahamsen, 1977) where the basement rocks are a southerly continuation of the Svecofennian zone in Sweden; remanence directions isolated in this study were interpreted to represent, in part, a Sveconorwegian overprint. Bylund (1981) studied dolerites and syenites intruded into the southern part of the Schistosity Zone in Sweden and reviewed palaeomagnetic data from the belt.

In the Rogaland Complex in the high-grade terrain of southwest Norway, Poorter (1972) reported a study from 3 sites in the Anorthosite Massifs, 9 sites in the Bjerkrem—Sogndal lopolith, and 4 sites in migmatites. Hargraves and Fish (1972), and Murthy and Deutsch (1974) have made investigations of the Egersund—Ogma Massif, with the former study including a petrologic investigation of the magnetic remanence carriers. A subsequent study of the farsundite complex by Murthy and Deutsch (1975) also incorporated a test for magnetic anisotropy.

Methodology of the present investigation

The study reported here has made a large-scale regional sample of the Sveconorwegian belt in 4 sectors spanning the width of the belt (Fig. 1). A

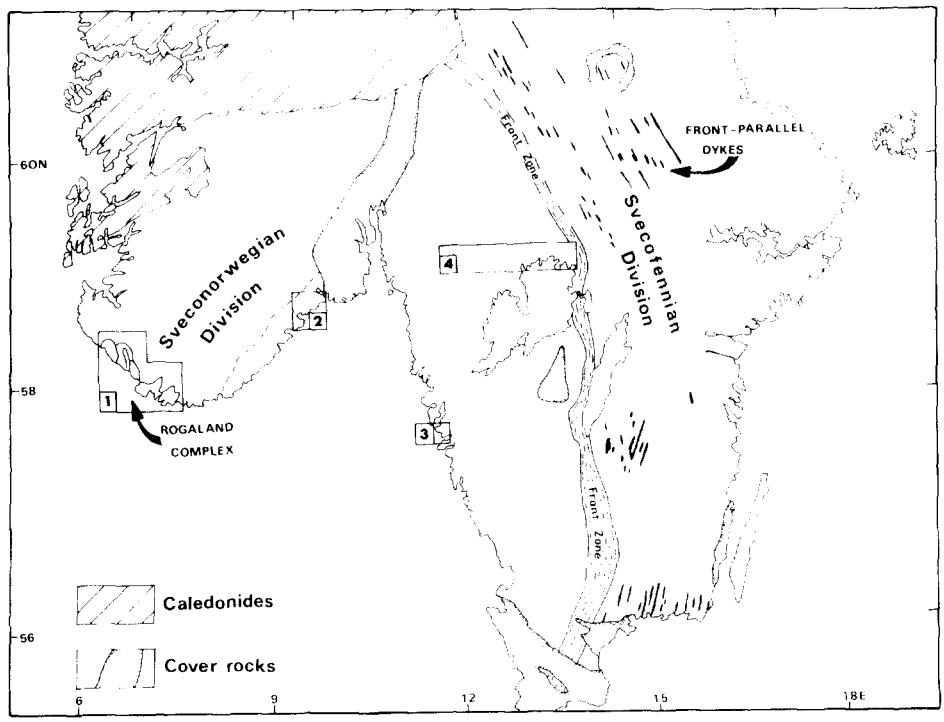


Fig. 1. Regional setting of the Sveconorwegian terrain in the Fennoscandian Shield with the locations (1—4) of the palaeomagnetic studies reported here. These locations refer to text figures 2 and 3, 8, 11 and 13, respectively.

large distribution of samples covers the high-grade terrains, widely intruded by anorthosite, and related complexes in southwest Norway (section 2) and the Bamble sector of southeast Norway (section 3); with more restricted collections from norite intrusions of the Tjörn area of western Sweden (section 4) and hyperite bodies up to the western margin of the Schistosity Zone between latitudes 59 and 59.5°N in Sweden (section 5).

A sampling site comprises a minimum of 7 cores distributed over several metres of outcrop and drilled with a portable drilling motor of the kind described by Doell and Cox (1967). Cores were orientated by sights on the Sun or, in the limited number of cases where this was not possible, by magnetic bearings in conjunction with topographic sights. The cores were sliced in the laboratory and measured with an automated parastatic magnetometer system. Alternating field (a.f.) demagnetisation is most suitable for analysing the broad range of coercivities present in the (predominantly magnetite-held) remanences in these rocks and was utilised in increments of 5 or 10 mT (milliTesla) as the standard method of treatment for this collection. Thermal demagnetisation of one or more samples from many sites was, however, also undertaken using a Schonsted thermal demagnetiser to test for components ineffectively resolved by the a.f. method, but the results of this study are

based largely on a.f. demagnetisation and may not have comprehensively recognised components identifiable by thermal cleaning only. The calculation of mean directions followed a method similar to that used by Dagley and Ade-Hall (1970), but it utilised specified intervals of the demagnetisation treatment as identified by study of demagnetisation plots and vector diagrams. Curie point determinations (saturation magnetisation, J_s, versus temperature) were also made to complement the analysis at many sites.

Theoretical aspects of the acquisition of magnetic remanence under the conditions of protracted uplift and cooling experienced by the Sveconorwegian terrain are discussed by a number of workers (Ueno et al., 1975; Morgan, 1976a; Pullaiah et al., 1975). The range of critical blocking temperatures is depressed below the range determined in the laboratory. Practical studies of Proterozoic terrains (Morgan, 1976b; Piper, 1981) emphasise the most characteristic feature of the palaeomagnetism in uplifted terrains, namely the resolution of a sequential migration of the palaeopole position in place of the single coherent directions isolated from more rapidly-cooled situations. This variation is present within single cores, between cores from the same site with different coercivity or blocking temperature spectra, and between areal distributions of sites, and this same feature is clear from regional assessments of palaeomagnetism in the Grenville terrain of the Laurentian Shield (McWilliams and Dunlop, 1978; Roy and Blythe-Robertson, 1979).

In principle, the acquisition of magnetisations in plutonic terrains should have been sufficiently slow to average out the effects of secular variation and, at any given crustal level, samples with higher blocking temperatures should record earlier parts of the a.p.w. path from those with lower blocking temperatures. In addition, if the range of critical blocking temperatures is distributed, demagnetisation should recover progressively higher blocking temperatures and, therefore, older segments of the total magnetisations so that sample directions move progressively back along the a.p.w. path (Morgan, 1976a, b). Although the coercivity spectrum cannot of course be directly linked to the blocking temperature spectrum, similar behaviours are observed with both a.f. and thermal techniques in samples with broad spectra (Morgan, 1976a, b; Piper, 1981) so that the phenomenon can, at least in some rocks, be recovered by both methods. Recognition of these effects requires that components illustrating them are shown to have a singular and primary origin (Piper, 1981).

THE SOUTH ROGALAND IGNEOUS COMPLEX AND FARSUNDITE, SVECONORWEGIAN TERRAIN, SOUTHWEST NORWAY

Outline of geology and age

In the Sveconorwegian terrain of southern Norway the metamorphic grade broadly increases radially outwards from greenschist facies in the

Telemark region to granulite facies on the southwest coast (South Rogaland) and the southeast coast (Bamble sector). The South Rogaland igneous complex (Fig. 2) is a ca. 1500 km^2 zone of intrusive bodies with discordant relationships with the surrounding migmatitic country rocks which have undergone granulite-facies metamorphism (Michot, 1960, 1969) grading downwards to amphibolite facies at the east of the complex (Hermans et al.,

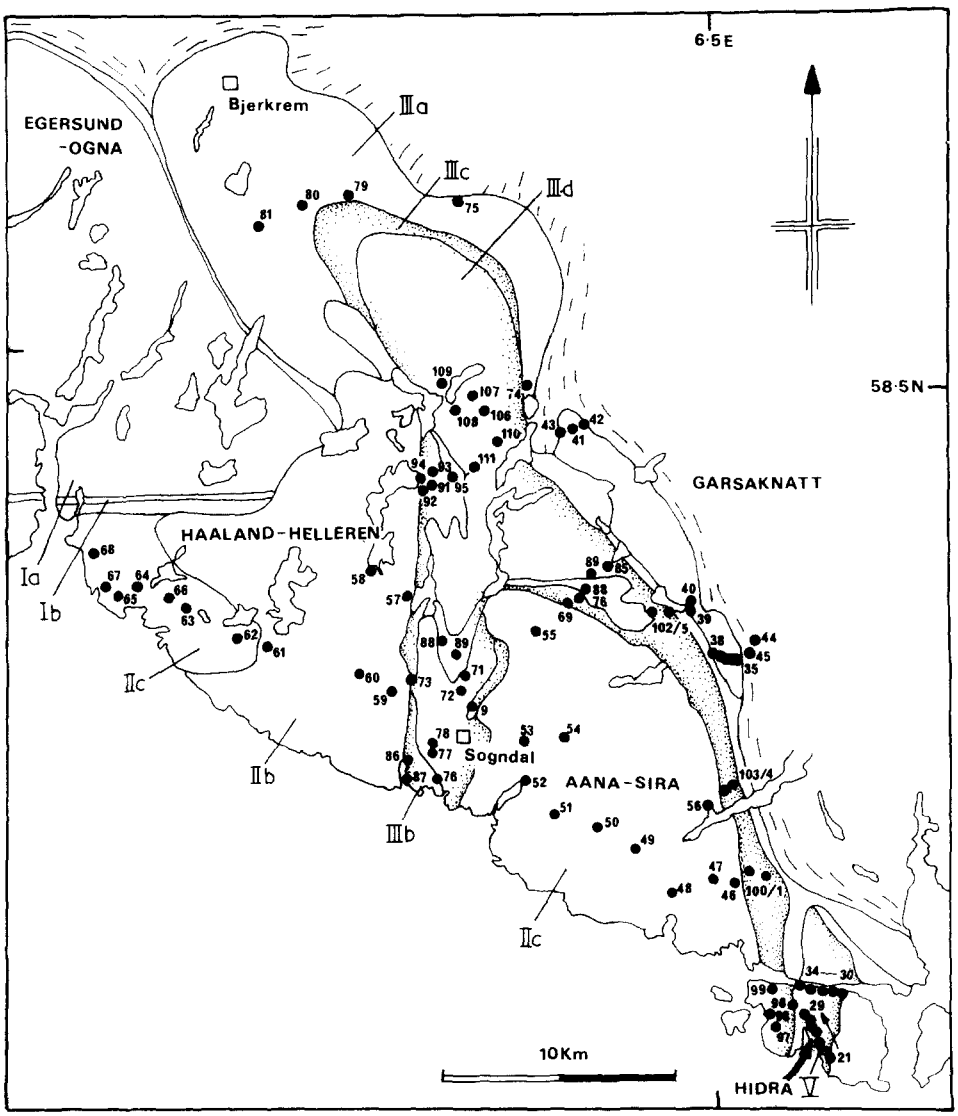


Fig. 2. Outcrop and sampling sites of the anorthosite and related intrusions within the Rogaland complex of southern Norway. Outline geology is after Michot (1960, 1969) and Hermans et al. (1975) and the large-scale time divisions recognised by these authors are given in Roman numerals (see text).

1975). First recognised as belonging to the anorthosite suite by Kolderup (1896), subsequent studies by Vogt (1924), Barth (1935) and P. Michot [see Michot (1969) for a bibliography] have contributed much detailed information on the lithology and origin of the complex.

The anorthosites are considered to have been intruded in a deep (25—35 km) catazonal environment (Michot, 1960, 1969; De Waard et al., 1974), and the emplacement of the various intrusive bodies can be related to 3 major phases of deformation which affected the regional metamorphic terrain (Michot, 1960; Hermans et al., 1975). The oldest intrusion is the Egersund—Ogna anorthosite body (Fig. 2), intruded during the first tectonic phase characterised by recumbent folding about N—S trending axes. This was followed by the marginal Lakssvesfjeld—Koldal noritic intrusion which separates the massif from migmatitic country rocks. The second phase is defined by folding along E—W axes when the rhythmically-layered Bjerkrem—Sogndal lopolith was intruded. During the third tectonic phase, defined by folding about vertical axes, the remaining monzonitic (the Eia—Rekefjord intrusion), monzonite and quartz-monzonitic (the Bjerkrem—Sogndal lopolith) units were emplaced (Hermans et al., 1975). Shortly after intrusion of the first unit of this body, the anorthosites and leuconorites of the Haaland—Helleren (Michot, 1960) and Aana-Sira massifs (Knorn and Krause, 1977) were intruded (Michot, 1969). The farsundite body was emplaced towards the end of the third tectonic phase (Pasteels et al., 1970) and the two outlying leuconoritic and anorthosite bodies of Garsaknatt and Hidra (Demaiffe et al., 1973) were intruded at some later time. Uplift and erosion of the crustal block occurred soon after this last event (Versteeve, 1975). The Egersund WNW dolerite dykes post-date all of these tectonic intrusive events, but are probably of Precambrian age (Versteeve, 1975).

The 420 km^2 farsundite body (Fig. 3) is a granitic body with anorthosite affinities, and this rock term, first used by Kolderup (1896), has subsequently been employed as a local specific name for the rocks of the Farsund area (Kolderup, 1935; Barth, 1935; Middlemost, 1968). Contact with the country rock gneisses and migmatites is only locally discordant and the farsundite was intruded in a deep plastic environment towards the end of the main deformational phase (Pasteels et al., 1970).

A considerable number of radiometric studies on the Rogaland rocks have been reported (Michot and Pasteels, 1968, 1969, 1972; Pasteels et al., 1970; Pasteels and Michot, 1975), and an overall synthesis has been presented by Versteeve (1975). Six zircons from the later phases of the Bjerkrem Sogndal lopolith gave a weighted mean $^{207}Pb/^{206}Pb$ age of 962 ± 40 Ma (Pasteels et al., 1970), recalculated to 945 ± 40 Ma by Versteeve (1975) and interpreted as the age of emplacement and beginning of solidification of the lopolith. The farsundite yields similar ages (977 ± 36 Ma on zircon and 944 ± 84 on sphene) and is thought to be of the same age as the lopolith (Pasteels et al., 1970). The granulite- to amphibolite-facies regional metamorphism reset most U—Pb and Rb—Sr whole-rock systems (Versteeve, 1975): the youngest

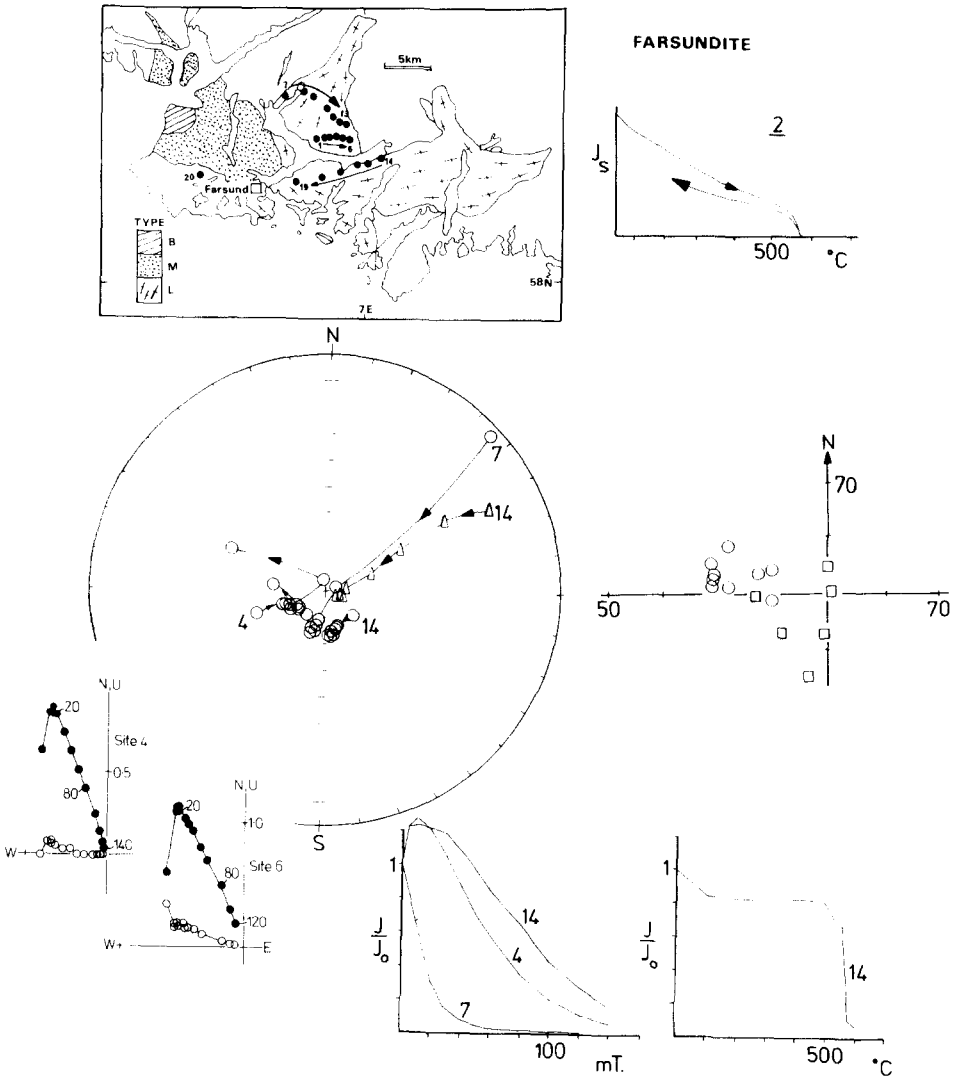


Fig. 3. Outline geology of the farsundite body after Middlemost (1968) with palaeomagnetic sampling sites. The facies of farsundite recognised by Middlemost (1968) are indicated in the key as L (leucofeldspathic), M (melanofeldspathic), and B (border micro granite/leucofeldspathic). In this and subsequent diagrams the directional behaviour of palaeomagnetic samples with progressive a.f. (circles) and thermal (triangles) treatment is shown on stereographic projections with closed symbols representing lower hemisphere plots. The decay of the natural remanent magnetisation (NRM) is shown as a fraction of the initial moment, J_0. A typical thermomagnetic determination (saturation magnetisation, J_s, versus temperature) is also shown; the J_s axis is arbitrary. The vector plots illustrate the initial removal of a low coercivity component followed by progressive subtraction of a single component; solid symbols are east and down, open symbols are east and north, figures are milliTesla. The expanded segment of an equal area plot shows the distribution of site mean directions from this study after a.f. and thermal cleaning.

Rb—Sr isochron ages obtained from the migmatites are 927 ± 26 Ma and 983 ± 88 Ma while ages of 900—850 Ma, characteristic of minerals in the migmatites, are interpreted as cooling ages fixed during post-tectonic uplift. Thus, it would appear that the later phases of the Bjerkrem—Sogndal lopolith were intruded shortly before commencement of uplift. The youngest (quartz-monzonitic) phase of the Bjerkrem—Sogndal lopolith yields an Rb—Sr isochron age of 842 ± 30 Ma (Versteeve, 1975) contrasting with the zircon age of 945 ± 40 Ma for the same unit. Versteeve interprets this younger age for the lopolith as that of final solidification of the quartz-monzonitic phase, and suggests that the difference between the Rb—Sr isochron ages obtained from the lopolith and the surrounding migmatites might be due to a higher partial pressure of the fluid phases of the lopolith delaying final solidification. The outlying Garsaknett body gave a single Rb—Sr mineral age on biotite of 902 ± 30 Ma consistent with mineral ages obtained from the migmatites.

Slow cooling of the igneous complex and the surrounding migmatites is implied by both the difference between the zircon and Rb—Sr isochron ages and by the different mineral ages. Versteeve (1975) dated minerals in samples of migmatite by both the K—Ar and Rb—Sr methods. The K—Ar ages on hornblende are some 50 Ma > Rb—Sr ages on biotite, which are in turn some 25 Ma > K—Ar ages on biotite; the systematic age sequence is K—Ar (hornblende) > Rb—Sr (biotite) > K—Ar (biotite) and spans some 75 Ma. Since argon is retained by biotite below temperatures between 150 and 250°C, and by hornblende at temperatures below ~ 500°C (Moorbath, 1967; O'Nions et al., 1969), and closure of the radiogenic strontium system in biotite takes place at ~ 300°C (Moorbath, 1967), the calculated rate of crustal cooling is ~ 4°C Ma^{-1}.

The farsundite complex

The farsundite complex is centred on the town of Farsund (58°5′N 6°48′E); sites numbered 1—6 and 8—13 were drilled within the northern limb of the intrusion and sites 14—19 sample the eastern and central parts of the complex (Fig. 3), with an additional site (20) sampling migmatite country rock. Total NRM's with scattered positive and negative inclinations rapidly converge with cleaning in fields > 10 mT to steep negative inclinations remaining stable to the limits of treatment. During this phase of higher cleaning treatment, directions tended to be not precisely static and sites 1—6, 11, 12 and 18 exhibited minor movements (mostly 3—4° over the a.f. cleaning range) towards an ESE direction. (Thermal samples illustrate a comparable systematic movement of the remanence vector over a range of cleaning fields and lose the bulk of this remanence close to the Curie point of magnetite.) The intensity plots illustrate an initial small rise followed by a progressive fall as a small lower coercivity and positive inclination component is removed during cleaning. Four sites have properties not typical of the remaining

group: 7 and 17 possess large low-coercivity components and are less stable to treatment, but converge in higher fields to directions typical of the remainder, while site 9 with similar properties converged to a shallow westerly direction (Table I).

The collective group mean directions from the northern part of the farsundite complex are elongated in an E—W direction which is close to the trend resolved by high field a.f. demagnetisation. This may represent the signature of an a.p.w. path recording during slow uplift and cooling similar to the example described by Morgan (1976b). The remaining sites from the central and southern parts of the complex have steeper inclination and are more dispersed. Murthy and Deutsch (1975) quote a pole from 7 sites in the farsundite, of which all but one come from the southern part of the complex. Although their calculation incorporates a 3.2° correction for the effects of anisotropy, it is not significantly different from the pole from the 6 sites of this study in the southern part of the complex. However, the pole from the 11 sites of this study in the northern limb of the complex is well defined and significantly different from both studies of the southern part of the complex (correction for a comparable degree of anisotropy would not affect this conclusion); the palaeopoles are interpreted in later sections.

TABLE I

Palaeomagnetic results for the farsundite complex, Sveconorwegian terrain, southwest Norway, after a.f. and thermal cleaning

Site no.	N	D	I	R	k	$J_0 \times 10^{-4}$
1	7	274.1	−69.8	6.96	168	1.2— 4.4
2	7	279.0	−68.9	6.97	234	2.0—10.8
3	7	294.8	−70.3	6.99	432	1.4— 8.9
4	7	284.1	−68.0	6.93	81	3.3— 6.8
5	7	273.1	−71.8	6.98	273	1.0— 5.4
6	6	276.7	−68.9	5.98	286	6.2— 9.1
7	8	236.9	−78.6	7.96	180	1.8—16.3
8	6	264.0	−80.2	5.99	541	4.4—14.5
9[a]	6	301.4	0.6	5.95	109	4.2— 7.4
10	6	—	—	—	—	3.3—71.3
11	7	292.4	−78.7	6.98	295	3.1—10.4
12	7	286.5	−76.6	6.99	516	3.6—24.0
13	6	272.7	−69.2	5.99	392	6.6—10.7
14	6	191.8	−83.8	5.99	404	4.5— 8.8
15	6	231.5	−79.0	5.99	517	3.1— 7.5
16	6	269.1	−77.3	5.99	551	1.9—10.6
17	6	194.9	−74.8	5.98	220	0.5—15.2
18	6	351.9	−83.5	5.99	389	0.9— 3.7
19	6	355.0	−88.5	5.97	164	1.2— 4.8
20	5	244.9	−71.2	4.98	167	0.7—11.5

D is the mean declination and I the mean inclination of N samples after cleaning with a resultant vector R and Fisher's precision parameter, $k = (N - 1)/(N - R)$. J_0 is the range of total NRM's in A.m^2 kg^{-1}. Site 20 is in migmatite country rock bordering the complex.
[a]Site mean directions excluded from the group mean direction.

The Hidra and Garsaknatt bodies

The central portion of the island of Hidra (58°13′N 6°35′E) is composed of leuconorite which, together with a continuation of the outcrop on the mainland immediately north of the island, forms the outlying Hidra body (Fig. 2). Samples were taken from 14 sites on the island, with 5 (30—34) on the north coast and the remainder (21—29) on the south coast, so that the total sample comprises two sections through the width of the body. A similar leuconorite outlier, northwest of Hidra, the Garsaknatt body was sampled at 9 sites forming a section across the width of the body.

On Hidra, sites from the north shore (which approach an anorthosite in composition) have scattered total NRM's which converge with a.f. cleaning to westerly declinations, with very steep negative inclinations (Table II) retained to the limits of a.f. treatment and the Curie point of magnetite (Fig. 4). Sites from the south shore are true leuconorites and exhibit generally similar behaviour although a number of lower coercivity cores yielded no meaningful data. Samples from two sites (22 and 27) initially yield steep negative westerly directions, but after cleaning in a field of 10—80 mT this component is removed in some samples to progressively recover a northerly direction with steep positive inclination approaching the only direction present at site 28 (Table II).

Of the 9 sites in the Garsaknatt body, 7 (35, 37, 38—41 and 43) yield very well-grouped directions with minimal a.f. cleaning, but during progressive cleaning all samples move systematically through a few degrees in an ESE

TABLE II

Palaeomagnetic results from the Hidra body, Sveconorwegian terrain, southwest Norway after cleaning

Site no.	*N*	*D*	*I*	R	*k*	$J_0 \times 10^{-4}$
21	4	282.0	−78.9	3.98	173	0.8 — 8.3
22	3	267.9	−58.0	3.00	474	0.2 — 0.9
23	5	335.0	−72.7	4.98	193	2.3 — 6.7
24	6	331.8	−78.5	5.94	80	3.1 —14.7
25	7	289.9	−70.9	5.92	6	1.3 —18.8
26[a]	5	67.8	−68.6	4.09	4	1.45— 4.5
27	5	307.3	−77.5	4.98	168	8.2 —71.2
28[a]	7	33.8	49.7	6.93	90	4.3 — 7.2
29	7	342.4	−81.5	6.17	7	0.9 —15.1
30	7	267.6	−67.1	6.99	879	1.3 — 2.5
31	6	292.3	−73.2	5.99	493	0.3 — 3.2
32	6	184.2	−86.2	5.82	28	2.1 —14.0
33	5	286.8	−69.6	4.81	21	0.4 — 2.0
34	6	295.0	−85.1	5.96	128	0.8 — 3.2

Symbols as for Table I.

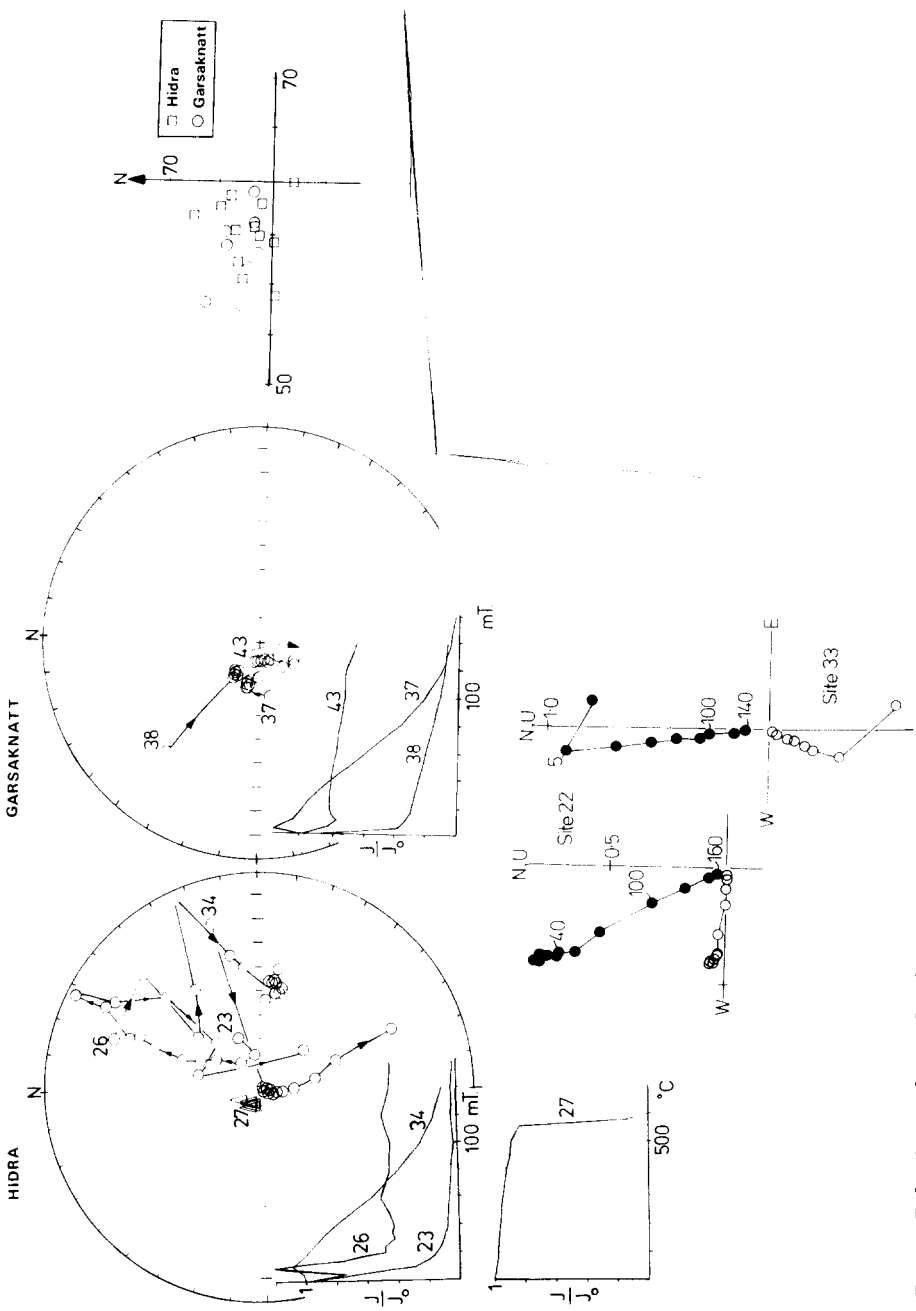


Fig. 4. Behaviour of samples from the Hidra and Garsaknatt bodies with progressive a.f. and thermal treatment, and a plot of the site group mean directions. The vector plots illustrate the subtraction of a single component by cleaning treatment. Symbols as for Fig. 3.

direction in a similar manner to the directions of some cores in the farsundite complex (Fig. 4). This movement is significantly removed from the great circle connecting the sample directions with the present geomagnetic field direction and appears to represent the recovery of progressively higher blocking temperature components acquired during slow cooling. In contrast to the characteristic steep easterly negative remanence present in most cores from this body, most site 42 samples migrate towards a shallower positive direction (Table III). In general, cores illustrate the initial increase in remanence followed by a progressive fall (Fig. 4) illustrating two divergent components with contrasting coercivities; the exception is site 43 where no decrease in moment takes place with a.f. treatment of a component which does not disappear until close to the Curie point of hematite. Two sites from migmatitic country rock adjacent to the eastern margin of the Garsaknatt body (44, 45) have magnetic properties comparable to those of the majority of sites from within the body (Table III).

TABLE III

Palaeomagnetic results from the Garsaknatt body, Sveconorwegian terrain, southwest Norway

Site no.	*N*	*D*	*I*	R	*k*	$J_0 \times 10^{-4}$
35	6	313.3	−76.5	5.99	595	6.3 — 8.4
36	6	332.2	−85.8	5.90	50	0.008— 1.3
37	6	285.7	−72.7	5.99	750	6.8 —13.3
38	6	293.9	−82.4	5.99	385	5.5 —18.5
39	6	279.8	−75.6	5.95	98	3.8 — 5.2
40	6	284.5	−62.5	5.92	59	3.3 —19.4
41	6	306.2	−74.5	5.96	125	0.4 — 0.5
42[a]	5	268.1	42.2	4.90	40	0.8 — 3.6
43	6	289.3	−62.7	5.95	95	0.03 — 0.09
44	7	275.5	−78.8	6.94	107	0.5 — 4.4
45	6	263.3	−64.0	5.94	86	0.2 — 0.5

Symbols as for Table I, sites 44 and 45 are from migmatite country rock.

The Aana-Sira Massif

This is a large complex composed of anorthosite and leuconorite centred on the town of Aana-Sira (58°18′N, 6°26′E). Sites 46—53 comprise a section through the southern part of the body, while sites 54, 55 and 56 are in the central, northern and north-eastern parts of the body, respectively (Fig. 2). The characteristic magnetisations are highly stable and directed near vertically upwards (Fig. 5). Samples retain this direction with no significant change in direction through the whole range of a.f. demagnetisation treatment (Table IV); this stable remanence typically resides in single components which remain largely unblocked until the Curie point of magnetite is reach-

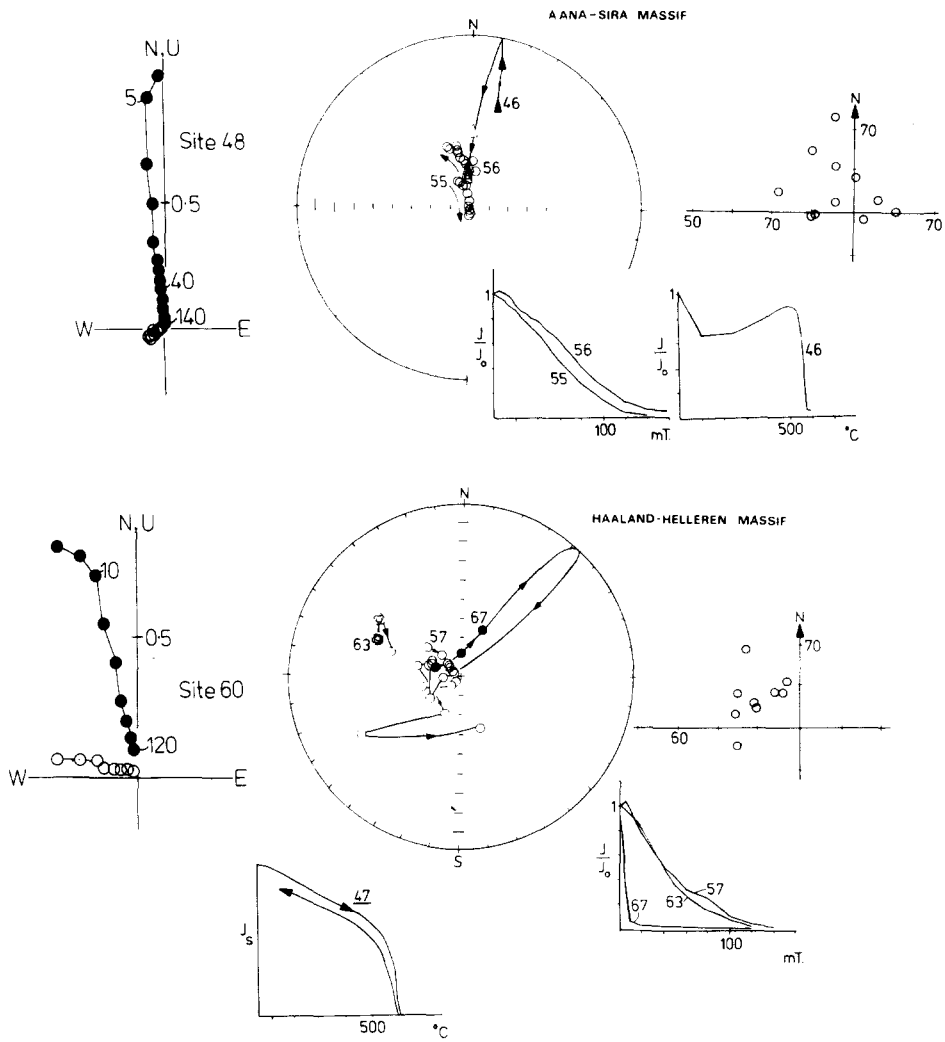


Fig. 5. Behaviour of some representative palaeomagnetic samples from the Aana-Sira and Haaland—Helleren Massifs with progressive a.f. and thermal treatment, with plots of the site group mean directions; a typical thermomagnetic determination is also illustrated. Vector plots again illustrate the subtraction of a single component.

ed (Fig. 5). The exceptions are sites 55 and 56 where the directions of magnetisation moved continuously through a few degrees towards less steep negative inclinations during progressive treatment, similar to many cores from the farsundite and Hidra bodies; in this instance the migrations are not systematic, however, and samples from the strongly-magnetised site 55 migrate to the SW quadrant while those from site 56 migrate to the NW quadrant.

TABLE IV

Palaeomagnetic results from the Aana-Sira Massif, Sveconorwegian terrain, southwest Norway

Site no.	N	D	I	R	k	$J_0 \times 10^{-4}$
46	6	3.6	−81.7	5.97	187	0.7— 6.5
47	6	283.6	−70.6	5.97	184	5.0— 1.1
48	6	264.6	−81.4	6.00	1211	2.9— 4.2
49	5	89.1	−80.3	4.98	256	5.2— 7.8
50	6	127.6	−86.8	5.97	181	3.0— 6.4
51	6	339.0	−78.2	5.99	798	6.6— 7.8
52	6	262.7	−80.0	5.93	73	1.8— 7.1
53	6	299.8	−84.9	5.99	517	1.1— 7.5
54	6	65.1	−83.3	5.98	232	2.2— 5.3
55	6	325.6	−71.9	5.98	225	3.4—24.1
56	6	349.0	−66.3	5.98	332	1.2— 1.8

Symbols as for Table I.

The Haaland—Helleren Massifs

The Helleren Massif, with similar petrologic characteristics to the Aana-Sira body, was sampled at 5 localities, three (59—61) along the main Sogndal to Egersund road and two (57, 58) further north in the central part of the intrusion. Still further to the west, bordering the coast between the Helleren and Egersund—Ogna bodies, lies the smaller Haaland anorthosite intrusion; 7 sites (62—68) are distributed along the length of this body (Fig. 2).

Magnetic properties are similar to the Aana-Sira Massif and cores from the majority of sites (57—62 and 65) were magnetised in WNW directions with very steep negative inclinations which remained stable to the limits of a.f. treatment; in general, there is no appreciable movement of the remanence vector during cleaning, although vectors from cores at sites 57 and 59 moved continuously through a few degrees of arc in a southerly direction. Two sites (67 and 68) with low coercivity components become less stable in higher a.f. fields, but yield comparable directions in low a.f. fields (Table V). Sites 63, 64 and 66 are magnetised with directions ~ 40° shallower and are excluded from the group mean calculation.

The Bjerkrem—Sogndal lopolith

This lopolith was intruded in 4 phases exposed around the margin of the Aana-Sira Massif and north of the Helleren and Egersund—Ogna bodies (Fig. 2). The earliest phase of anorthosite and norite is exposed in 3 areas. The larger two, in the vicinity of Bjerkrem (58°36'N 6°5'E) and Sogndal, were sampled at sites 74, 75, 79—81 and at sites 71—73, 76, respectively. The

TABLE V

Palaeomagnetic results from the Haaland—Helleren Massifs, Sveconorwegian terrain, southwest Norway

Site no.	N	D	I	R	k	$J_0 \times 10^{-4}$	
57	6	335.6	−80.6	6.00	1310	18.7	−51.1
58	6	325.6	−66.6	5.99	540	0.2	− 2.3
59	6	292.5	−78.7	5.99	665	3.6	−12.8
60	6	346.1	−78.9	5.98	231	3.7	− 6.1
61	6	298.0	−77.7	5.98	200	11.7	−16.8
62	6	299.1	−73.4	5.97	155	0.9	− 5.3
63[a]	6	294.7	−40.1	5.97	171	0.09	− 1.2
64[a]	6	282.3	−42.0	5.28	7	4.3	−12.6
65	5	327.2	−79.5	4.99	557	0.006	− 0.6
66[a]	6	306.2	−37.2	6.00	1145	5.0	− 8.7
67	7	253.5	−73.6	6.82	34	8.3	−28.8
68	6	280.8	−74.3	4.89	5	0.008	− 1.3

Symbols as for Table I.

third is a small exposure ~ 8 km northeast of Sogndal, sampled at sites 69 and 70. The second intrusive phase is a monzonite (the Eia—Rekefjord intrusion) exposed around Eia (58°27′N 6°17′E, sites 91—95), south of Eia along the western margin of the lopolith (sites 86 and 87) and as a small intrusion on the eastern side of the lopolith 2 km north-north-east of Sogndal (site 90). The third intrusive phase of the lopolith, also a monzonite, is exposed along the northern limb 7 km southeast of Bjerkrem, in the southern limb 4 km north of Sogndal (sites 88 and 89), and in the eastern limb 9 km northeast of Sogndal (sites 82—85). The fourth and youngest phase is quartz-monzonite and comprises the uppermost layer of the complex exposed in the centre and along the eastern border of the Aana-Sira Massif. Ten sites were sampled in this eastern limb of the outcrop, four (96—99) on Hidra, two each (100, 101 and 103, 104) traversing the outcrop west of Flekkefjord, and two sites (102, 105) further north between the eastern exposure of the first phase of the lopolith and the Garsaknatt body. An additional 6 sites (106—111) are located in the northern part of the outcrop (Fig. 2).

Most samples from the anorthosites, leuconorites and norites of the first intrusive phase possess a remanence residing in at least two constituents: a large unstable lower coercivity component (insignificant at sites 77 and 78) is largely removed in low a.f. fields while a weak high-coercivity, and probably single (Fig. 6), component has a magnitude little affected by subsequent treatment, although some directional changes continue to take place such that optimum groupings of directions are achieved with intermediate demagnetising fields. Less steep negative inclinations, a minor component of the other bodies, are present at sites 71 and 81.

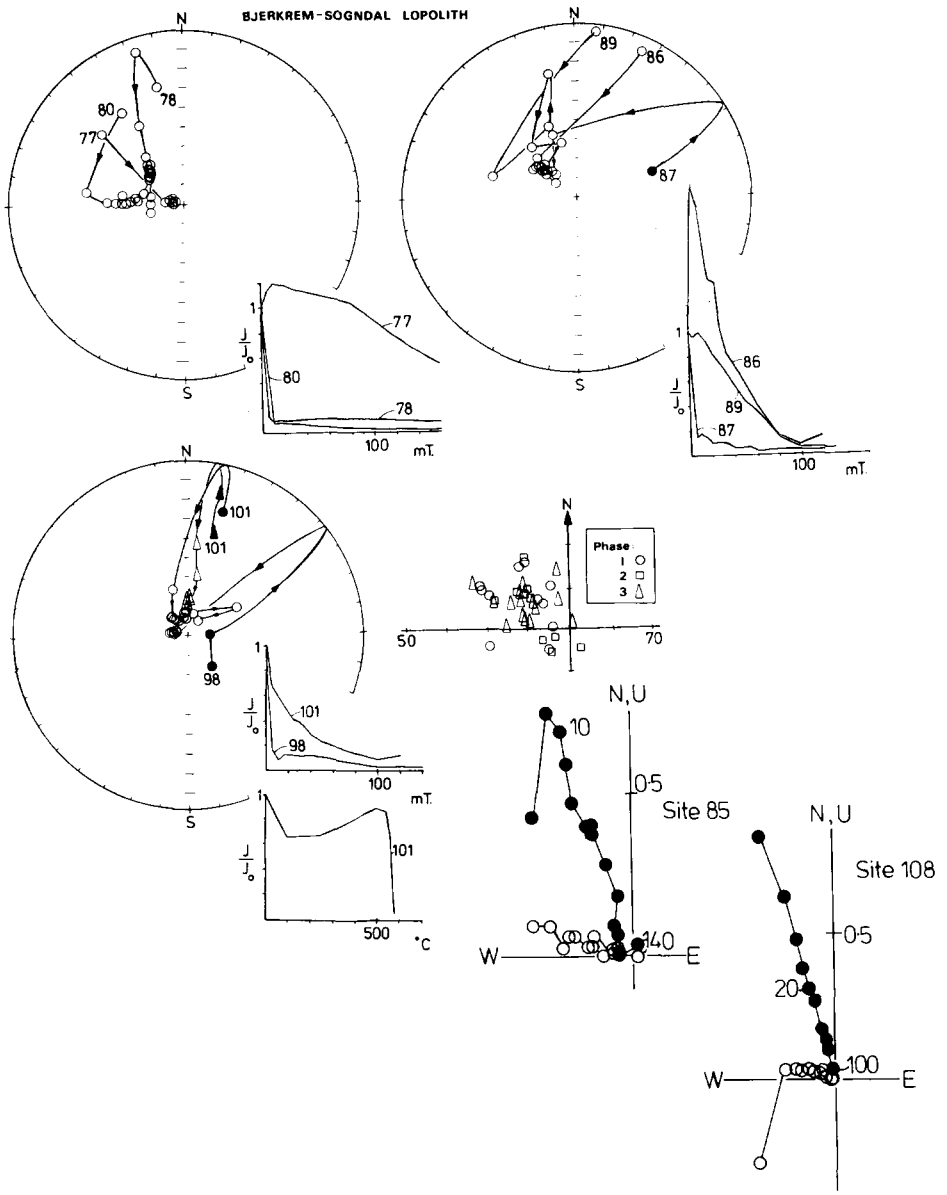


Fig. 6. Behaviour of samples from the Bjerkrem—Sogndal lopolith with a.f. and thermal treatment, and site group mean directions after cleaning. The vector plots illustrate the single component nature of the bulk of the remanence. The expanded equal area plot shows that the vectors isolated from the first (1), second and third (2) and fourth (3) phases of the lopolith are comparable.

TABLE VI

Palaeomagnetic results from the four intrusive phases of the Bjerkrem—Sogndal lopolith, Sveconorwegian terrain, southwest Norway

Site no.	N	D	I	R	k	$J_0 \times 10^{-4}$
First phase						
69	6	259.0	−69.6	5.88	40	3.6—13.6
70	6	225.7	−83.1	5.87	40	12.9—23.6
71[a]	6	98.7	−54.0	5.89	47	7.2—11.2
72	6	292.0	−68.7	5.43	9	5.7—13.7
73	6	336.1	−78.8	5.88	42	3.4—14.6
74	6	327.4	−70.5	5.93	75	5.0—29.9
75	6	321.0	−71.1	5.95	104	14.3—36.2
76	6	313.0	−82.2	5.91	55	2.5— 5.0
77	6	277.9	−85.7	6.00	1495	1.9— 9.8
78	4	294.2	−66.8	3.95	59	4.4—22.7
79	6	311.5	−80.3	5.73	19	6.6—16.4
80	6	294.8	−65.8	5.95	102	5.2—11.3
81[a]	6	267.9	−46.9	5.90	49	8.2—16.3
Second and third phases						
82	6	246.1	−83.0	5.93	75	0.6— 2.0
83	6	151.0	−83.6	5.91	54	0.8— 6.5
84[a]	7	186.0	−36.2	4.56	2	0.1— 3.0
85	6	289.8	−71.2	5.99	393	0.4— 1.1
86	6	311.4	−76.9	5.96	121	0.6— 7.1
87	6	327.7	−69.5	5.97	155	4.5— 8.8
88	7	231.0	−85.9	6.69	19	1.7— 7.4
89	6	305.5	−74.6	5.98	201	1.5— 4.1
90	6	218.5	−82.7	5.64	14	1.1— 6.1
91[a]	6	308.1	−53.5	5.92	59	0.4— 4.8
92	6	307.8	−77.5	5.53	11	1.6— 3.0
93[a]	6	306.8	−49.0	5.91	55	1.7— 6.8
94[a]	6	300.2	−41.5	5.51	10	3.2— 6.1
95	6	309.0	−76.7	5.79	23	1.0— 1.9
Fourth intrusive phase						
96	6	271.4	−75.3	5.97	181	1.3— 5.2
97[a]	6	300.8	59.5	5.68	16	4.3— 9.3
98	6	297.0	−81.0	5.96	128	1.2— 3.6
99	6	299.0	−77.2	5.89	45	1.3— 3.9
100	6	275.9	−80.9	5.97	191	1.8—17.7
101	6	337.1	−83.2	5.98	279	2.3— 6.0
102	6	282.1	−77.8	5.66	15	1.3—12.2
103	5	40.3	−89.2	4.97	143	1.0— 1.8
104	6	303.4	−78.5	5.96	129	1.5— 4.3
105	6	311.1	−75.3	5.96	116	1.3— 6.2
106	6	294.4	−64.4	5.89	46	0.2— 1.1
107	6	305.3	−75.6	5.93	69	0.9— 3.6
108	6	289.3	−70.8	5.99	412	1.9— 6.8
109	7	292.4	−74.7	6.93	90	0.2— 0.9
110	6	348.7	−76.3	5.62	13	0.3— 1.5
111	6	278.6	−78.5	5.94	83	0.2— 1.6

Symbols as for Table I.

Most samples from the second and third monzonite phases of the lopolith also possess a large low-coercivity component, usually of steep positive inclination, which is removed in low a.f. fields, sometimes with an initial increase in intensity, to recover the characteristic steep negative direction present as the minor high coercivity component; in general there is again an optimum cleaning field with the directional behaviour becoming less stable in the highest a.f. fields. Two sites (85, 89) fail to show the low coercivity component while 3 sites (91, 93 and 94) yield the NW direction with an inclination ~30° shallower than the majority of sites.

Samples from the fourth (quartz-monzonite) phase possess more stable remanence than the remaining phases; 3 sites (98, 102, 110) hold appreciable low coercivity components and site 97 is dominated by this component yielding no systematic high coercivity remanence. Samples from the majority of sites (96, 98—101, 103—109 and 108) yield NW directions with steep negative inclinations which move progressively through a few degrees in a SE direction during the whole range of a.f. treatment. The site mean directions from the Bjerkrem—Sogndal lopolith are listed in Table VI and plotted on Fig. 6.

Nature and carriers of magnetisation in the Sveconorwegian intrusive complexes

Following intrusion of the igneous bodies forming the South Rogaland complex, a period of slow cooling resulted from crustal uplift (Versteeve, 1975). Crystallisation of the feldspars, in which much or all of the stable remanence appears to reside (see below), took place at temperatures of at least 660°C (Michot, 1969; Hargraves and Fish, 1972). The high stability to a.f. and thermal cleaning, the identification of predominantly single-component behaviour by cleaning treatment, the consistency of the palaeomagnetic results over the whole plutonic complex and the uniformity of K—Ar ages all suggest that the original TRM acquired during uplift has not been significantly affected by later events. In addition, there is no isotopic evidence for later metamorphism or for thermal effects related to the Caledonian orogeny.

Thermomagnetic analyses (saturation magnetisation, J_s, versus temperature) identify single reversible Curie points, attributable to magnetite, between 550 and 590°C in most samples; a lower Curie point at ~300°C in some samples is eliminated by heating and may be due to pyrrhotite (Figs. 3, 5 and 6). It is unlikely that these determinations are completely representative of the magnetic mineral carriers in these rocks because magnetite has a saturation magnetisation some 218 times greater than hematite and the latter may escape detection. By petrologic investigation, Hargraves and Fish (1972) identify hematite—ilmenite intergrowths forming fine rod-shaped inclusions within the plagioclase crystals of the anorthosite and hematite—ilmenite intergrowths as an accessory mineral; in the monzonites, approximately

equal proportions of magnetite and ilmenite were identified as discrete crystals or coarse intergrowths. Magnetisations of high stability in plutonic rocks have been widely linked to fine, needle or blade-shaped inclusions of magnetite or hematite exsolved along cleavage planes in silicates (Evans and McElhinny, 1966; Hargraves and Fish, 1975; Piper, 1979). In somewhat analogous slowly-cooled plutonic basic rocks of West Greenland, Morgan and Smith (1981) link low coercivity remanence, eliminated at the initial phases of a.f. treatment, to large homogeneous magnetite grains and high coercivity remanence to submicron magnetite inclusions in plagioclase. We have no direct petrologic evidence bearing on the magnetic carriers in these rocks, but it is generally true in the Rogaland rocks that while directional behaviour is highly systematic, susceptibilities and total NRM intensities range by 5 orders of magnitude and appear to reflect the wide variation in magnetic mineral content: because samples, both with and without discrete magnetite, exhibit similar magnetic intensities after cleaning in fields of 5—10 mT, because of the resistance to a.f. cleaning and the high unblocking temperature of the stable remanence, the high coercivity stable remanence is linked to magnetite and/or hematite inclusions in the silicate phases, while the low coercivity remanence at some sites is provisionally linked to large magnetite and/or pyrrhotite grains.

Palaeomagnetic poles from the South Rogaland complex

It has been noted that the site mean directions from the farsundite complex vary with geographic position: the shallower inclination sites lie north of Lyngdalsfjorden and the others lie to the south; mean palaeomagnetic poles have, accordingly, been calculated separately for these two groups of sites. The individual site mean directions from the various intrusive units of the South Rogaland complex tend to have elongated distributions (Figs. 3—6) and the collective mean palaeomagnetic poles lie along a short swathe of just over 30° of arc incorporating the two poles from the farsundite complex at each end (Fig. 7). The pole from the Aana-Sira Massif lies at one end of the swathe, poles from the Haaland—Helleren Massifs and Bjerkrem—Sogndal lopolith lie in the middle and the poles from the Hidra and Garsaknatt bodies lie near the other end. Thus, there is a systematic trend across the basement complex, with the exception being the Aana-Sira Massif where sample blocking temperature spectra are consistently higher than those from elsewhere. The only significant exception from the calculated group means are a few sites with similar declinations, but systematically shallower inclinations than the remainder. There is no continuous distribution between this group of 5 sites (91, 93, 94, 63 and 66) and a separate group mean is calculated as pole 'Y' in Table VII.

Previous palaeomagnetic studies of the South Rogaland complex were concentrated in the northwest part of the complex, and the two outlying bodies of Hidra and Garsaknatt have not previously been sampled. With the

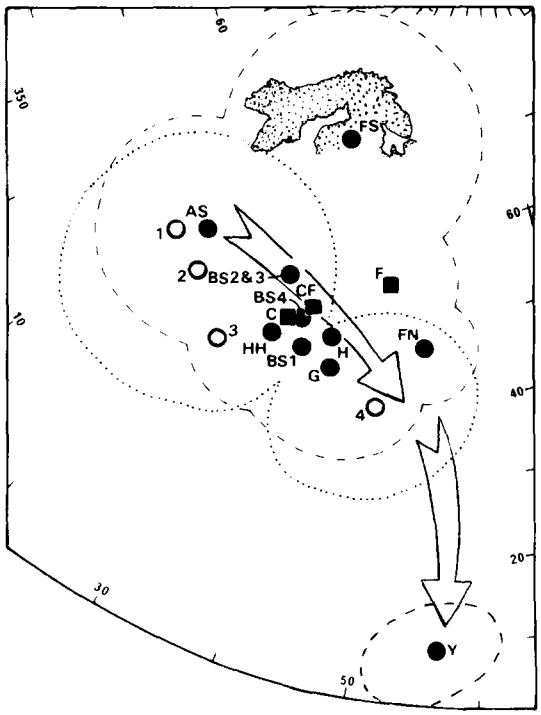


Fig. 7. Palaeomagnetic poles from the constituent bodies of the Rogaland complex plotted on a Lambert Equal Area projection. The (reversed) palaeopoles plotting nearest to the sample location are shown here and the direction of palaeofield migration inferred from demagnetisation trends, the regional outcrop and blocking temperature spectra are shown by arrows. Poles numbered 1—4 and plotted as open circles refer to previous palaeomagnetic studies of the Rogaland complex. All these poles are referenced in Table VII.

exception of the study of the Egersund—Ogna Massif by Murthy and Deutsch (1975), these studies have yielded poles 1—4 in Table VII representative of the Rogaland area as a whole; they are precisely incorporated within the swathe defined by the present study (Fig. 7).

Age, rate and direction of movement of the a.p.w. path recorded by the South Rogaland complex

Under conditions of slow uplift-related cooling it is predicted that the effective blocking temperature ranges of the magnetic grains will be lowered and broadened because of the time-dependent viscous properties of the magnetisation (Ueno et al., 1975; Pullaiah et al., 1975). It appears, however, that this effect has been greatly over-estimated in earlier calculations because it failed to accommodate the change in activation energies of magnetic grains (Dodson and McClelland-Brown, 1980). Even under the slow cooling rates evident in this area it appears that the effective blocking temperature range

TABLE VII

Summary of mean palaeomagnetic directions and poles from major intrusive complexes in the Sveconorwegian terrain of southwest Norway

Rock unit	No. of sites	D	I	R	α_{95}	k	Palaeomagnetic pole		(dp, dm) or (α_{95})	
							°N	°E		
This study										
Farsundite (F)	17	270.8	−77.6	16.78	9.3	72	−50.7	226.5	(7.4, 7.9)	
northern part (FN)	11	277.3	−73.2	10.94	3.5	174	−43.9	232.2	(5.6, 6.2)	
southern part (FS)	6	233.7	−84.3	5.94	7.5	81	−63.4	207.8	(14.5, 14.7)	
Hidra body (H)	12	291.8	−76.6	11.80	5.8	56	−43.1	219.7	(10.1, 10.8)	
Garsaknatt body (G)	8	294.9	−74.5	7.91	6.2	80	−39.6	221.4	(10.2, 11.3)	
Aana-Sira Massif (AS)	11	326.5	−83.0	10.80	6.5	51	−46.3	197.4	(12.4, 12.6)	
Haaland–Helleren Massifs (HH)	9	304.7	−77.4	8.92	5.2	99	−41.1	212.5	(9.1, 9.7)	
Bjerkrem–Sogndal lopolith										
Phase 1 (BS1)	11	298.8	−76.3	10.85	5.7	65	−40.9	216.9	(9.7, 10.5)	
Phases 2 and 3 (BS2 and 3)	10	296.1	−80.8	9.87	5.8	71	−47.6	210.6	(10.7, 11.1)	
Phase 4 (BS4)	15	297.6	−78.0	14.89	3.4	128	−43.5	215.0	(6.0, 6.4)	
Anorthosite complex (C)	76	299.5	−78.3	74.94	2.0	70	−43.5	213.7	(3.5, 3.7)	
Anorthosite complex and farsundite (CF)	93	294.1	−78.4	91.64	1.8	68	−44.9	215.8	(3.2, 3.4)	
Sites 63, 66, 91, 93, 94 (Y)	5	302.9	−44.4	4.96	7.5	106	−6.8	235.7	(5.9, 9.4)	
Previous studies										
Farsundite body, southern part [M and D, 1975 (1)]	7	341	−82		5	142	−43	194	(9, 10)	
Egersund–Ogna and Haaland–Helleren Massifs [H and F, 1972 (2)]	5	315	−82		8	92	−42	200		(15)

TABLE VII (*continued*)

Rock unit	No. of sites	*D*	*I*	R	α_{95}	*k*	Palaeomagnetic pole °N	Palaeomagnetic pole °E	(*dp*, *dm*) or (α_{95})
Egersund–Ogna Massif [M and D, 1974 (3)]	26	—	—		—	—	−37	207	(8, 8)
Bjerkrem–Sogndal lopolith [P, 1972 (4)]	9	283	−73		8	45			
Anorthosite Massifs [P, 1972 (4)]	3	318	−58		38	12	−36	227	(10, 12)
Migmatites [P, 1972 (4)]	7	286	−70		11	29			

H and F = Hargraves and Fish, P = Poorter and, M and D = Murthy and Deutsch. Where the palaeomagnetic pole was calculated from virtual geomagnetic poles, the radius of the circle of confidence about the pole at the 95% probability level is given. The angles *dp* and *dm* are the semi-axes of the oval of confidence about the pole at the 95% probability level, in the colatitude direction and perpendicular to it, respectively. *N* is the number of sites. For an explanation of the other symbols used refer to Table I.

will only have been lowered marginally below those ranges determined in the laboratory. However, it is certain that the cooling periods of individual samples through their blocking temperature spectra were sufficiently slow to average the components of secular variation (Morgan, 1976a). The magnetisations of these samples were generally unblocking over temperature ranges of at least 10°C which corresponds to a cooling period of several Ma; the effects of short events of opposite polarity may also be averaged out although it is implicit in the preservation of a predominant magnetisation that the polarity of the field was strongly biased in one direction (Morgan, 1976a).

The third effect of slow cooling is that a.p.w. motions may be recorded within individual samples provided that the rate of a.p.w. was sufficiently rapid and the blocking temperature range is sufficiently broad. The site mean directions from this study form elongate distributions (Figs. 3—6) with palaeopoles which approximately parallel the swathe of group mean directions, and cores from the two intrusive units showing the lowest ranges of blocking temperatures (the Garsaknatt body and the northern part of the farsundite body) exhibit progressive movement SSE or parallel to this swathe. As both a.f. and thermal demagnetisation have the resultant effect of selectively removing the magnetisations residing in grains with lower blocking temperatures, this movement may indicate a migration from younger to older palaeopoles. This interpretation is suggested by the observations that the great circles defined by the movement incorporate neither the present field direction nor the predominant Caledonian palaeofield direction, and it is essentially confirmed by the distribution of high to low blocking temperature components: the blocking temperature of the rocks of the Aana-Sira Massif, as defined by the thermal demagnetisation studies, are higher than the other intrusive units of the South Rogaland complex. In addition, the blocking temperature spectra of rocks from the southern part of the farsundite body are higher than those within the northern limb of the intrusion. In both cases the implied direction of a.p.w. is away from the sampling locality (Fig. 7).

Since the rate of cooling of the migmatites surrounding the igneous complex is approximately known, as are the blocking temperatures of the rock samples, it is possible to estimate the rate of a.p.w. represented by the swathe in Fig. 7. Samples from the farsundite body yielding mean unblocking temperatures of 520 and 570°C have palaeopoles separated by an angular distance of 35° and imply an a.p.w. rate of 3° Ma^{-1}. In the Aana-Sira Massif a set of samples with mean unblocking temperature of 590°C yield a pole separated by 8° from a set of samples in the Bjerkrem—Sogndal lopolith with a mean unblocking temperature of 565°, implying an a.p.w. rate of 1.3° Ma^{-1}. These estimates are appreciably higher than most Phanerozoic rates, but appear to be fairly typical of the Phanerozoic era (Piper, 1981, 1982).

Thus, the collective poles of this study which define a swathe of ~30° in length appear to represent 10—20 Ma of a.p.w. movement. The width of the distribution is presumed to reflect real complexities of polar movement

during this interval because the components of secular variation should have been averaged out. Although we have been unable to correct for shape anisotropy, the effect of this is believed to be small for two reasons: in the farsundite body with strong visible anisotropy the greatest apparent deflection of the remanent vector by this effect was found to be 5.5° and to average 3.2° (Murthy and Deutsch, 1975). Secondly, the rocks of the Bjerkrem—Sogndal lopolith are folded, but no significant correlation has been found here between the direction of dip of the layers of the lopolith and the direction of magnetisation.

The palaeopoles from the farsundite body are slightly displaced from the almost parallel swathe defined by the anorthosite complex (Fig. 7). This is one discrepancy which may have a tectonic explanation because the farsundite lies adjacent to the North Sea margin: a southerly tilting of ~10° along the coastal margin here could accommodate the difference between the magnetic vectors in this body and the anorthosite complexes.

The mean 'Y' direction computed from the 5 aberrant sites with shallow inclinations is considered to be an extension of the a.p.w. path because the magnetic components relating to these poles have unblocking temperatures mostly between 400 and 450°C contrasting with the unblocking temperatures of 500—580°C in the remaining stable material. The present study has not succeeded in identifying the intervening path in the rocks of this region.

The migmatites west of the Bjerkrem—Sogndal lopolith have K—Ar ages on hornblende of 891 ± 30 Ma and 890 ± 30 Ma, Rb—Sr ages on biotite between 903 ± 30 and 850 ± 20 Ma, and K—Ar ages on biotite of 823 ± 25 and 812 ± 25 Ma (Versteeve, 1975). The mineral ages correspond to approximate cooling temperatures between 500 and 200°C, and the magnetic components, with the possible exception of the 'Y' magnetisation, would thus have been fixed by these times. A maximum age is represented by the 1000—945 Ma zircon ages which appear to represent the age of emplacement, but the 845 ± 30 Ma Rb—Sr whole rock isochron age on the final two intrusive stages of the lopolith (Versteeve, 1975) is not compatible with the migmatite data. Versteeve reconciles this anomalous age by arguing that strontium isotopes remained homogeneous in the lopolith for a long period after intrusion. The consensus of 14 mineral ages from the migmatites is that the magnetisations represented by the 25° swathe in Fig. 7 have represented some 10—20 Ma of a.p.w. movement between ca. 940 and 900 Ma; the 'Y' magnetisation may be as young as 840 Ma.

BASIC INTRUSIONS OF THE BAMBLE SECTOR, SVECONORWEGIAN TERRAIN, SOUTHEAST NORWAY

Outline of geology and geochronology

The Bamble sector of the Sveconorwegian belt, lying along the southeast coast of Norway between Kragero in the east and Kristiansand in the west, is

a high-grade metamorphic basement (Bugge, 1943; Barth and Dons, 1960; Barth and Reitan, 1963) bounded to the north by the Palaeozoic rocks of the Oslo graben and to the west by the Kristiansand–Porsgrunn fault where the lower-grade metamorphic basement of Telemark is downfaulted. The Bamble sector comprises migmatites, gneisses, and high-grade metasediments intruded by syn- to post-tectonic granites and basic rocks (hyperites); the latter comprise both norites and gabbros some of which are olivine-rich.

From some 50 K–Ar determinations O'Nions et al. (1969) concluded that a thermal maximum of 700–650°C occurred at 1080 Ma and was immediately followed by a long period of cooling with temperatures of 200–150°C being reached by 950 Ma. They concluded that the gabbroic bodies were emplaced in two stages, the first at around 1090 Ma and the second at around 1010 Ma. A Rb–Sr whole-rock isochron age of 1139 ± 100 Ma (O'Nions and Baadsgaard, 1971) was interpreted to represent the thermal maximum and accords closely with the K–Ar data. More recent studies regard this age as indicating a re-homogenisation of the isotopic systems by a granite–pegmatite intrusion or shearing in rocks originally formed prior to 1500 Ma (Field and Raheim, 1981). Since the thermal maximum occurred some 200–150 Ma before the final stages of cooling recorded by K–Ar mineral ages at closure temperatures of 200–150°C, a cooling rate of ~3°C Ma^{-1} is applicable to this part of the Sveconorwegian belt and comparable to the Rogaland area.

Palaeomagnetic results from the Bamble sector

This survey has concentrated on the bodies of gabbroic composition intruding the northern part of the Bamble sector between 9.3 and 9.6°E and 58.8 and 59.1°N near Kragero. The palaeomagnetic results are tabulated area by area in Table VIII and site localities are shown in Fig. 8; sites 138 and 152 are dykes cutting the gabbro and norite intrusions.

Samples from the majority of sites have moderate to low coercivities and the typical pattern of behaviour with a.f. treatment is a rapid removal of a variable component near the present field. In the Nenset, Gumöy and Langöy rocks (Table VIII) this is reflected in a directional change from variable positive inclinations to NW, intermediate to steep, negative inclinations and an initial increase in remanence in the higher coercivity samples.

In some samples this is followed by a much smaller systematic migration in a contrasting direction at higher stages of treatment (site 128, listed in Table VIII at initial and end stages of treatment) and these components are removed by thermal cleaning between 560 and 600°C (Fig. 9). Two sites at Nenset and one on Gumöy with very low coercive force spectra (114 and 120) yielded no stable end points. Considering the data from the Nenset, Gumöy and Langöy intrusions as a whole, two clear sets of magnetisations are present (Fig. 10). The predominant one is NW negative and site 121 on Gumöy is nearly antiparallel to this group, while a subordinate group has

TABLE VIII

Palaeomagnetic results from basic intrusions in the Bamble sector Sveconorwegian terrain, Bamble area, southwest Norway

Site no.		N	D	I	R	k	$J_0 \times 10^{-5}$
Nenset							
112		6	4.3	−39.5	5.95	103	13 — 81
113		6	321.2	−39.1	5.92	64	13 —101
114		6	—	—	—	—	67 —443
115		6	332.9	−48.2	5.97	174	35 —334
116		6	337.7	−32.6	5.81	26	24 —201
117		6	322.9	−39.2	5.52	10	0.2 — 4
118		6	324.4	−40.1	5.90	51	35 —244
119		6	319.4	−46.8	5.84	32	65 —209
120		6	—	—	—	—	0.2 — 14
Gumöy							
121		6	131.2	22.5	5.34	8	0.05— 6
122		6	203.3	−81.3	5.98	202	2 — 13
123		6	299.3	−70.8	5.81	26	3 — 24
124		6	330.6	−47.9	5.35	8	44 —122
125		6	237.2	−30.0	4.54	3	0.02— 0.2
126		6	210.5	−40.3	5.19	6	0.03— 0.06
Langöy							
127		7	184.8	−36.0	6.92	71	6 — 43
128	(i)	7	195.2	−38.4	6.98	305	22 — 57
	(ii)	7	181.6	−29.0	6.94	93	
129		6	192.6	−43.1	5.97	144	7 — 43
130		7	348.2	−28.3	6.88	48	2 — 17
131		7	104.4	4.2	6.43	11	0.01— 0.03
132		6	343.9	−40.8	5.75	20	10 — 36
133		7	324.6	−33.5	6.77	26	0.5 — 4
Dördäl district							
134	(i)	7	236.5	−34.4	6.86	43	1 — 11
	(ii)	3	334.6	−46.1	2.92	24	
135	(i)	5	223.8	−30.6	4.78	18	0.2 — 34
	(ii)	3	341.6	−26.8	2.91	22	
136		6	297.2	−58.2	5.89	44	0.1 — 44
137	(i)	5	315.9	−56.7	4.89	38	0.2 — 49
	(ii)	2	185.2	−55.5	1.99	148	
138		7	343.3	−37.3	6.91	68	36 — 70
139		7	359.2	−46.9	6.99	511	18 — 32
140		5	31.3	−61.4	4.78	18	8 — 22
141		7	327.0	−50.3	6.98	273	29 — 64
142		6	331.8	−45.6	5.95	102	0.1 — 49
143		6	223.0	−31.9	5.99	1109	4.3 — 11
144		6	329.7	−48.2	5.86	35	2 — 21
145		5	140.6	49.6	4.76	17	0.1 — 1
146		7	222.3	−33.2	6.45	11	<0.1
147		6	195.1	−16.3	5.92	62	<0.3
148		3	338.3	−34.3	2.98	88	3 — 14
149		5	308.5	−52.8	4.92	50	16 — 79
150		6	326.7	−45.4	5.85	34	4 — 39
151		4	321.9	−45.1	3.94	54	0.1 — 13
Kragero district							
152		4	178.5	−38.6	3.96	74	2 — 6
153		5	128.3	40.7	4.93	58	4 — 41
154		4	127.1	64.4	3.93	42	0.3 — 14
155		6	323.4	−37.6	4.74	4	7 — 10

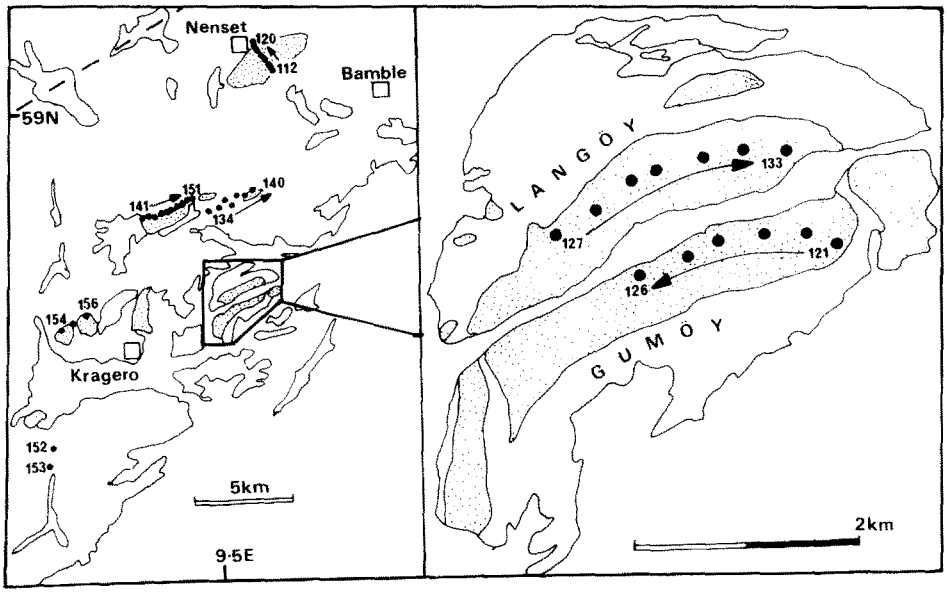


Fig. 8. Palaeomagnetic sampling sites and outline geology in the Bamble sector of south-east Norway. The NE—SW major fault in the west of the map upfaults this high-grade terrain against the low-grade terrain of the Telemark region.

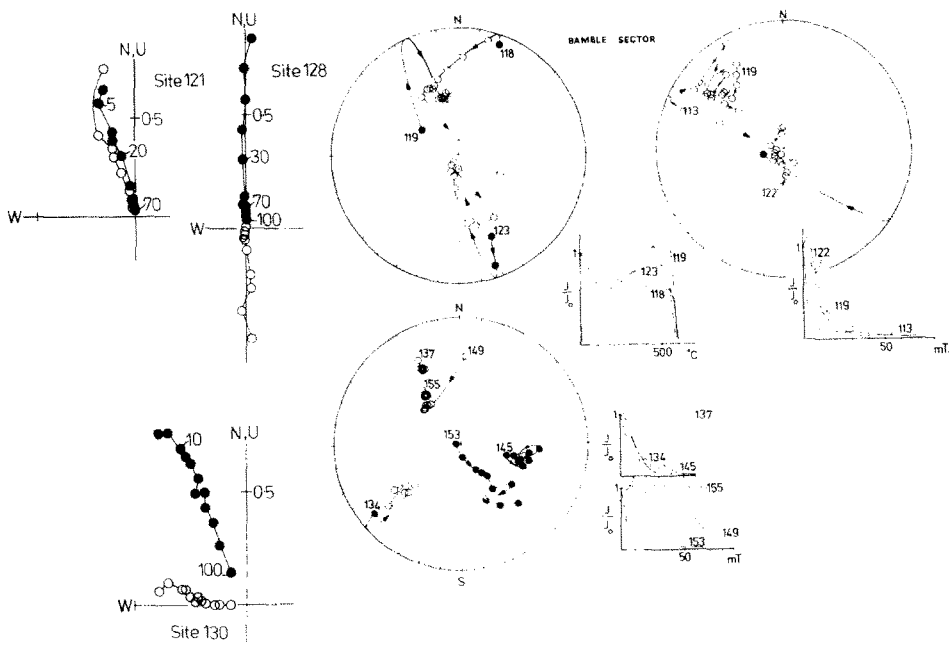


Fig. 9. Palaeomagnetic behaviour of some representative sites in the Bamble sector with a.f. and thermal treatment. Vector plots illustrate the apparent single component nature of the contrasting directions recognised in these bodies.

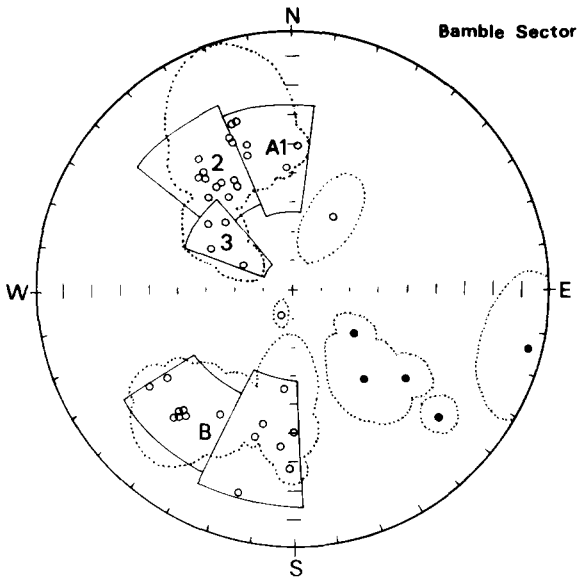


Fig. 10. Site mean directions from the Bamble sector. The NNW directed 'A' magnetisations mostly reside in hemo-ilmenite components and are complemented by a minor group of antiparallel directions in lower blocking temperature directions. The subsidiary 'B' magnetisations appear to be magnetite-held and mostly have lower blocking temperatures than the 'A' group although there are some exceptions. The sites have been grouped into approximately 30 × 30° segments of the stereonet as shown to compute the representative palaeopoles of Table IX.

SSW negative directions. From these collections it is not possible to make precise generalisations about the two groups; the subordinate group tends to include the samples with weaker magnetisations and less inherent stability, but there are important exceptions, notably sites 128 and 129.

More information relevant to these magnetisations is derived from the Dördäl District (Table VIII) where the samples from the bodies in the east of this district yielding SSW negative directions all have low coercivities with moments falling exponentially with a.f. treatment and eliminated by thermal cleaning close to the Curie Point of magnetite (Fig. 9). By contrast, all samples in this area falling in the NW negative group have high coercivities and weak total NRM's, and show little or no fall with progressive a.f. treatment; these directions are retained here above the Curie Point of magnetite and in some instances (sites 134, 135 and 137, Table VIII) the low coercivity component can be completely removed to recover the high coercivity component. This rule is also applicable to the basic intrusion northwest of Dördäl, except that some of the NW negative directions reside in lower coercivity components that illustrate progressive fall with a.f. treatment; this is notably the case in the coarse gabbro facies at sites 148—150, and site 145 illustrates an antiparallel component also in low coercivity material. The NW

TABLE IX

Mean palaeomagnetic poles from the Bamble sector of the Sveconorwegian terrain

Pole ref.	Site mean directions included in calculation	*N*	*D*	*I*	R	*k*	Palaeomagnetic pole		
							°E	°N	*dp/dm*
A3	123, 136, 137, 149, 154[a], TPA	6	305.7	−64.1	5.91	55	227.5	−23.9	11.6/14.6
A2	113, 115, 117, 118, 119, 121[a], 124, 133, 134, 141, 142, 144, 145[a], 148, 150, 151, 153[a], 155, TPK	19	325.1	−43.4	18.74	69	220.5	1.0	3.2/5.1
A1	112, 116, 130, 132, 135, 138, 139, TPB, TPC/TPB, TPE, TPL[a]	11	348.7	−42.6	10.73	37	199.6	5.9	5.6/9.4
B1	125, 126, 134, 135, 143, 146	6	225.3	−35.7	5.92	65	309.8	−39.2	5.6/9.7
B2	127, 128(i), 128(ii), 129, 137, 147, 152	7	187.8	−36.9	6.84	38	357.7	−51.3	6.8/11.7

[a]Sites reversely magnetised with respect to the remainder.
Lettered site references are given by Poorter (1975).

negative component is also found in the Kragero district (Table VIII) in high coercivity remanence, but 2 sites here have reversed magnetisations (154 has dual polarities) in lower coercivity magnetisations that are eliminated close to the Curie Point of magnetite.

Collectively, the basic bodies in the Bamble sector define two principal groups of directions (Fig. 10). The majority group with NW negative magnetisations (A) is complemented by a subsidiary group of antiparallel directions at sites 121, 145, 153 and 154 with lower coercive force and blocking temperature spectra. These directions fall along a swathe between D = 342, I = +28 and D = 298, I = +71, but there is as yet no clear evidence defining unequivocally the palaeofield migration represented by this path because the high coercivity components, presumably residing in hematite or a hemo-ilmenite, are distributed along the swathe. The subordinate SSW negative group (B) appears to reside entirely in lower coercivity magnetite-held components and systematic migration is observed with high field treatment in an anticlockwise direction at site 128.

Poorter (1975) also resolved single component directions of magnetisation, one nearly antiparallel to the remainder, by a.f. and thermal treatment, from 6 sites in hyperites and amphibolites of the Bamble sector; they all belong to the predominant NW negative group and lie along the distributions of these site mean directions resolved from this study (Fig. 10). To compute representative group mean directions, the site mean directions are grouped into approximately 30 × 30° sectors of the stereonet, as shown in Fig. 10 and listed in Table IX. We regard these palaeopoles as representative of the magnetisations acquired during uplift and cooling following the peak of regional metamorphism at 1140—1080 Ma, contemporaneous with, or younger than, the episodes of intrusion of the basic bodies (1080—1010 Ma), and appreciably older than the later stages of regional cooling dated at 950 Ma.

THE NORITE INTRUSIONS OF BRATTÖN AND ÄLGÖN, SVECONORWEGIAN TERRAIN, WEST SWEDEN

Geology and age

The islands of Brattön and Älgön lie at 57°55′N 11°48′E on the west coast of Sweden, 25 km north-north-west of Gothenburg. They are composed of norite with anorthosite inclusions intruded into gneisses and schists (Fig. 11). First mentioned by Svedmark (1888), they were briefly described by Ljungher (1927) and have been the subject of detailed study by Bergström (1961). Much of the norite is unaltered, but in parts of the bodies pyroxenes have altered to hornblende and feldspar has altered to sericite. The inclusions of anorthosite range from blocks 15 m in size down to single phenocrysts and are believed to be the remnants of a crust formed at the top of the noritic magma chamber, but which shattered at a later stage of the intrusion

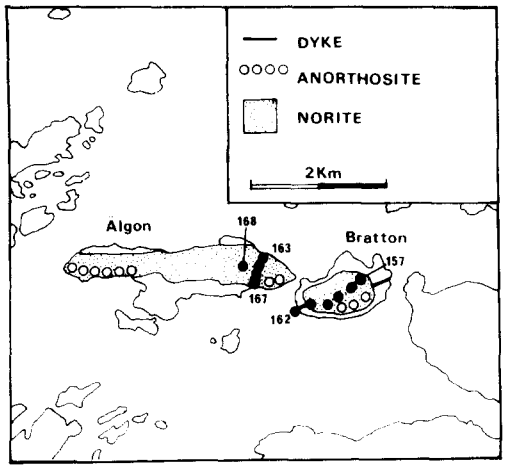


Fig. 11. Outline geology and distribution of sampling sites on the islands of Brattön and Älgön in the Tjarnes area of western Sweden.

(Bergström, 1961). The contacts are distinct with some brecciation, and the absence of grain size diminution towards the margin suggests that the body was intruded into country rock at elevated temperatures. Two dolerite dykes, possibly related to the norite, extend outwards from the intrusion on Brattön.

The geological evidence suggests that these bodies were intruded during the final stages of the last tectono—thermal episode to affect these rocks (Bergström, 1961). Mica schists from this region have yielded K—Ar ages of 1030 and 1005 Ma while this general area of Sveconorwegian basement yields K—Ar ages in the range 1091—909 Ma (Magnusson, 1960; Boch, 1964) interpreted to represent closure of the mineral systems during uplift and cooling. The late- to post-tectonic Bohus granite, 80 km north of Älgön, yields K—Ar ages of 977 and 880 Ma (Magnusson, 1960) and 880—830 Ma (Skiöld, 1976), in addition to a Rb—Sr isochron of 891 ± 35 Ma (Skiöld, 1976). Skiöld also obtained a Rb—Sr isochron age of 1027 ± 40 Ma and K—Ar ages of 1000—800 Ma on the Dal supracrustal group which lies in the Sveconorwegian zone, and from these and earlier dates he concluded that emplacement of the Bohus granite took place during the post-tectonic uplift in this area at 1000—800 Ma and marked the final phase of igneous activity. The norite bodies on Brattön and Älgön were probably intruded after the climax of the second Sveconorwegian metamorphic event (1000—950 Ma) which affected the whole of the Sveconorwegian basement (Skiöld, 1976; O'Nions and Baardsgaard, 1977) and before the final stages of uplift and cooling recorded by the K—Ar mineral ages. The dolerite dykes were apparently intruded later because they incorporate devitrified glass.

On the island of Brattön, 5 sites (156—160) comprise a section through the body of an altered norite, with the exception of one core in an anorthosite block (156-3), and the WSW dyke was sampled at site 161 (Fig. 11). Six further sites (162—167) sample norite from the Algön body.

Palaeomagnetic results

Total NRM's are mostly directed north westerly with shallow negative inclinations although a minority of sites (157, 159 and 160) with intensities an order or more higher than the remainder have steep positive inclinations. With progressive cleaning, samples migrate progressively through small angles to increasingly negative inclinations (sometimes accompanied by initial increase in intensity, Fig. 12) and within-site grouping generally improves at higher fields. These magnetisations disappear close to the Curie point of

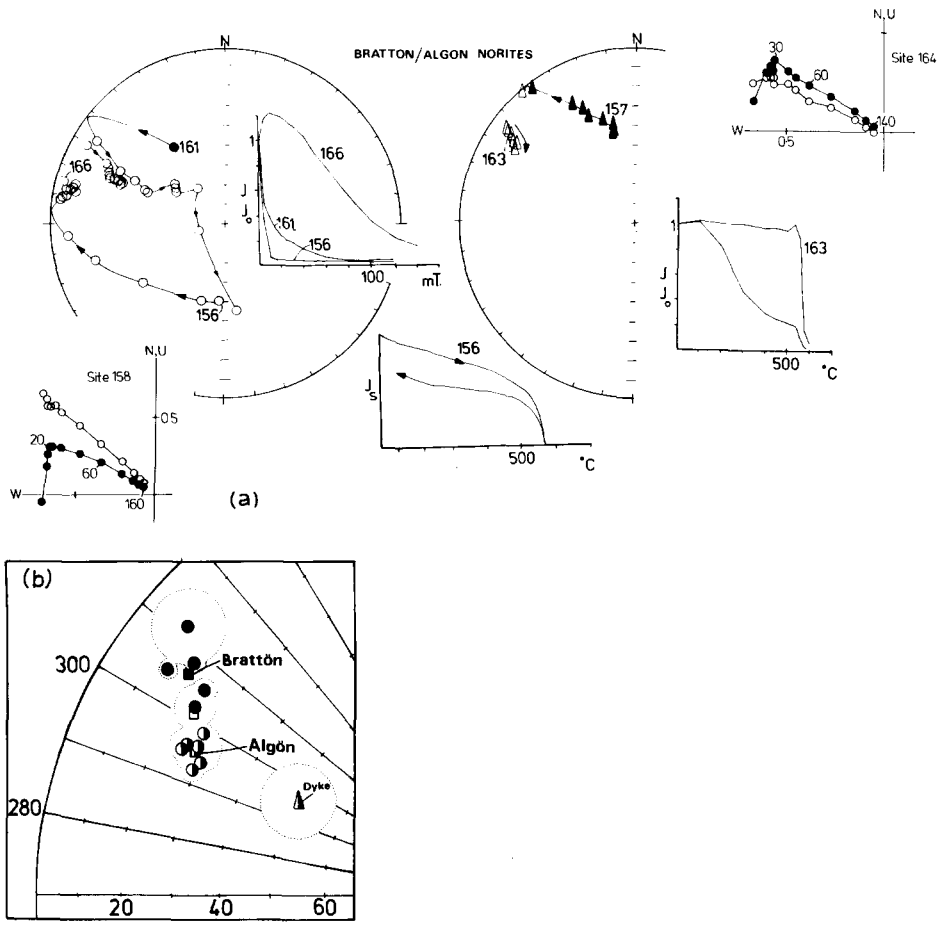


Fig. 12. (a) Behaviour of some palaeomagnetic samples from the Brattön and Älgön norite bodies with progressive a.f. and thermal treatment, and a typical thermomagnetic determination. The representative vector plots illustrate the resultant effect of the cleaning treatment. (b) The distribution of site and group mean directions from the Brattön and Älgön norites after cleaning treatment.

magnetite (Fig. 12) and three strongly-magnetised sites (156, 157, 159) conform to the behaviour of the remainder at higher stages of treatment. The dolerite dyke (161) migrated to steeper negative directions than the norite with progressive treatment; although coercivities were higher, within-site grouping deteriorated markedly with demagnetisation at ~40 mT, which yields the mean listed in Table X.

Bergström (1961) identifies large (0.3—0.6 mm) titanomagnetite grains, and some pyrite and ilmenite grains without exsolution textures; in addition, the plagioclase crystals contain fine needles. The thermomagnetic determinations yield a main Curie point at 550—570°C, but the presence of a second magnetic phase is identified by a tail with low saturation remanence and a Curie point of 580—600°C. In the context of studies in comparable plutonic rocks (e.g., Hargraves and Fish, 1975; Morgan and Smith, 1981) it seems likely that the component with high saturation remanence is to be linked to the large titanomagnetite grains which hold components eliminated in low a.f. fields, and the low saturation remanence is to be linked to a stable high coercivity titano—hematite exsolution phase in the silicates.

TABLE X

Palaeomagnetic results from the islands of Brattön and Älgön, Sveconorwegian terrain, southwest Sweden, after cleaning

Site no.	N	D	I	R	k	$J_0 \times 10^{-4}$
Brattön						
156	5	301.8	−24.3	4.99	307	0.7—209
157	7	304.8	−23.6	6.99	546	11 —729
158	6	307.6	−18.3	6.00	1111	6.0— 11
159	6	304.5	−14.0	6.00	1167	17 —452
160	6	311.8	−11.9	5.93	76	15 — 18
Dyke						
161	5	296.4	−51.3	4.96	100	1.8— 2.8
Älgön						
162	6	296.3	−28.2	5.97	189	16 — 19
163	6	298.5	−28.3	5.99	894	15 — 26
164	6	292.3	−29.5	5.99	884	19 — 23
165	6	294.9	−25.5	6.00	1114	15 — 24
166	5	293.7	−30.7	6.00	1166	9.7— 13
167	6	295.6	−26.1	5.99	334	11 — 16

Brattön and Älgön norite—anorthosite complex mean

					Pole	(*dp*, *dm*)
11	300.4	−23.8	10.89	89	5°N 249°E	(3, 5)

Symbols as for Table I. The palaeopole for the Brattön body (N = 5, α_{95} = 6.3°) is at 246°E 10°N (dp = 3°, dm = 7°) and the palaeopole for the Älgön body (N = 6, α_{95} = 2.3°) is at 249°E 5°N (dp = 3°, dm = 5°).

With the exception of the dolerite dyke, the site mean directions all lie on a short swathe (Fig. 12), with those from Brattön lying on the northern part and those from Algön lying on the southern part. The site means from Algön are tightly grouped, but those from Brattön are distributed along the swathe; this directional change is progressive across the island of Brattön (Fig. 11, Table X). The swathe incorporates some, but not all (the directional change of some samples lies along a great circle incorporating the present field direction) of the small migration during progressive cleaning. Since the complex was intruded into the basement at elevated temperatures and cooled during slow uplift, we link these magnetisations to the mineral ages lying between 1130 and 909 Ma, but clustering in the interval 1047 and 978 Ma. Skiöld (1976) also concludes that this sector of the Sveconorwegian basement closed to argon diffusion at ~1000 Ma and the primary remanence of the norite bodies would appear to be of about this age. The palaeopole from the dolerite (243°E 14°S) will be younger than this and can only be regarded as a virtual geomagnetic pole.

MINOR DOLERITE (HYPERITE) INTRUSIONS IN THE SVECONORWEGIAN TERRAIN IN WESTERN SWEDEN

Outline geology

In common with the Bamble sector, the Sveconorwegian terrain in western Sweden is extensively invaded by basic intrusions. In Sweden, these intrusions are mostly small lensoid bodies a few metres in width intruded parallel to the regional foliation. They are sparsely distributed through much of the terrain, but are very densely intruded towards, and into, the Schistosity Zone where they range from unaltered dykes with chilled margins to completely amphibolitised bodies in localised shear zones subjected to intense deformation. The intrusions of the Kristineham area (palaeomagnetic sites 175—180) have been described in detail by Wiman (1968); they include olivine-bearing ophitic dolerites containing orthopyroxene, ilmenite and magnetite. There is typically a transition from massive olivine-bearing centres into amphibolitic marginal zones within the individual bodies, such that the ophitic texture is gradually destroyed and the pyroxenes and the olivine are replaced by amphibole.

The age of hyperite intrusion is not well known, indeed it is possible that they were intruded during several intervals. Mulder (1971) derived K—Ar ages in the range 1569—786 Ma and Klingspar (1976) dated hyperite dolerites from areas further south in the range 1595—679 with a peak of K—Ar ages between 1050 and 750; Rb—Sr whole-rock determination on a hyperite yielded an age of 1565 ± 230 Ma, although at Karlshamn they intrude granite dated 1422 ± 31 Ma. The concentration of K—Ar ages suggests that uplift and cooling of the terrain incorporating these dolerites was related to stabilisation of the Sveconorwegian belt, but they may have been intruded, in part, some time before this.

Palaeomagnetic results

The palaeomagnetic samples from 14 sites include a number of isolated bodies on, or close to, the E18 road plus 6 bodies in the intensely-intruded zone close to the Sveconorwegian margin (Fig. 13).

The most characteristic behaviour of these rocks with progressive a.f. and thermal treatment is the retention of very steep positive or negative directions to the limits of treatment (169 and 178, Fig. 14); in some instances small progressive migrations take place (174) which are non-systematic within sites. In some samples there is removal of a lower coercivity component with shallow inclination (176), and at site 177 an optimum grouping in

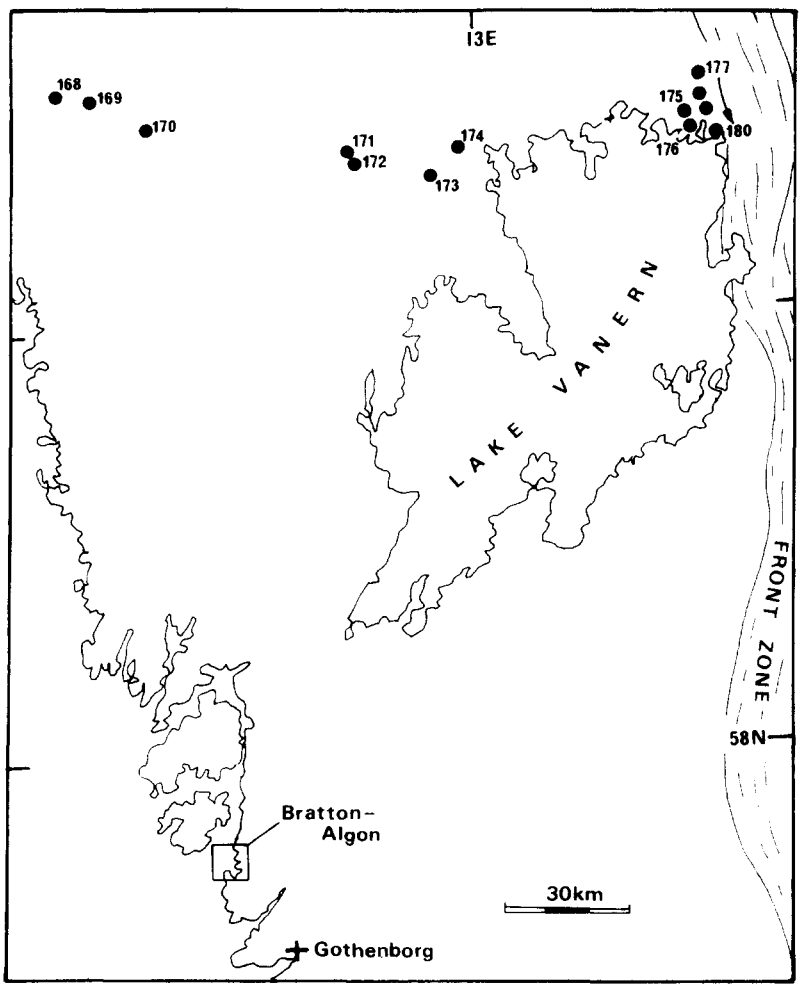


Fig. 13. Distribution of sampling sites in hyperite bodies in the eastern part of the Sveconorwegian terrain.

low fields (Table XI) is progressively destroyed in higher fields as some samples migrate to the steep negative direction; cleaning was unsuccessful in removing a smeared distribution and no mean is listed for the high blocking temperature component here. The sites with contrasting shallow southerly

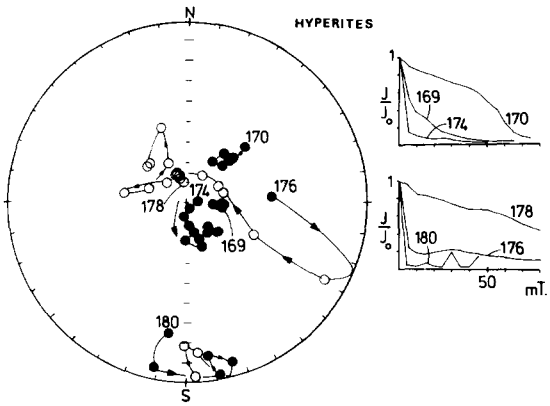


Fig. 14. Behaviour of some palaeomagnetic samples from hyperite bodies with progressive a.f. treatment.

TABLE XI

Palaeomagnetic results from minor dolerite (hyperite) intrusions in the Sveconorwegian terrain of western Sweden

Site no.	*N*	*D*	*I*	R	*k*	J_0 ($\times 10^{-5}$ A.m^2 kg^{-1})
168	8	152.0	74.1	7.74	27	0.4 — 4.6
169	7	108.5	71.9	6.97	210	9.8 —142.7
170	6	40.8	69.3	5.94	81	0.6 — 2.5
171	5	94.4	66.3	4.93	60	0.1 — 4.9
172[b]	7	82.4	27.9	6.90	60	0.04— 5.4
173[c]	7	169.8	−5.9	6.57	14	0.4 — 8.1
174	8	174.4	80.7	7.92	87	11.0 — 42.8
175	—	—	—	—	—	0.3 — 3.8
176	6	332.9	−82.7	5.87	39	10.3 — 71.7
177[b]	6	106.4	2.3	5.70	17	2.2 — 31.4
178	6	327.5	−77.3	5.94	81	2.1 — 30.3
179[c]	5	178.0	−7.6	4.82	22	0.04— 0.3
180[c]	9	198.9	−11.3	8.28	11	8.3 — 16.8

Group mean (sites 168, 169, 170, 171, 174, 176, 178, MZI, MZM)

					Pole	(*dp*, *dm*)
9	123.0	77.7	8.84	49	42.7°N 41°E	(13.1, 13.9)

Symbols as for Table I.
[b]Excluded from group mean calculation.
[c]Low coercivity component, may be Palaeozoic in age.

directions (173, 179 and 180) all have low coercivity and low within-site precisions; these directions merit little interpretation and may be of Palaeozoic age.

The site mean directions are of dual polarity and compare closely with 2 sites (M2M and MZI) of the earlier study by Mulder (1971). The probability that other ages of magnetisation are represented in the hyperite bodies is suggested by 3 other sites of Mulder (MZK, MZL and MZS) and 2 sites (172 and 177) of this study [see also discussion by Bylund (1981)]. No change in site mean direction is obvious towards the Schistosity Zone and a larger sample would be needed to resolve this. The mean direction of all sites with inclinations $> 60°$ is $D = 243$, $I = -78$, and compares closely with the mean direction of $D = 287.2$, $I = -83.8$ derived from 14 sites in hyperites and syenites from the southern outcrop of the Schistosity Zone in Scania (Bylund, 1981).

1050—850 Ma A.P.W. PATH OF THE FENNOSCANDIAN SHIELD AND TECTONIC IMPLICATIONS

The Sveconorwegian a.p.w. loop

The palaeopoles resulting from this study are regarded as post-tectonic in age and date either from the times of Sveconorwegian intrusion or the uplift and cooling of the terrain documented by the ubiquitous K—Ar ages falling in the range 1100—800 Ma. This is clearly the case in the Bamble and Rogaland areas where intrusive ages in this range define the maximum possible ages of the magnetisations; it is less clear in Sweden where the Brattön and Älgön poles can only be linked in a general way with the regional cooling history and the hyperites are not well dated. Even the limited data from the country rocks given here and by Poorter (1975) are unlikely to record a pre-Sveconorwegian remanence, both because the prevailing upper amphibolite- to granulite-facies conditions imply that rocks were at temperatures in excess of 600°C for a protracted period, and because the data from the country rocks accord with those from the Sveconorwegian intrusions.

As previously analysed the Rogaland complex records a ca. 30° motion of the pole away from the Fennoscandian Shield at ca. 940—870 Ma. The pole from the hyperites of this study falls near the younger end and the pole from the Scanian hyperites (Bylund, 1981) falls near the centre of this swathe (Fig. 15). Their magnetisations are almost certainly of this age and, indeed, the systematic migration from steeper to shallower inclinations recorded in a hyperite body by Mulder (1971) is explicable in the context of the Rogaland data. In view of the limited evidence (Mulder, 1971; Klingspar, 1976) that intrusion of these bodies may predate 1000 Ma, the question of whether the hyperites record a metamorphic magnetisation imposed during uplift of the front zone, or an intrusive magnetisation, must remain an open one pending further geochronological studies. The preservation of both

polarities within this suite could not be accommodated by very slow cooling and suggests that relatively short magnetisation events are recorded in these units (Mulder, 1971; Poorter, 1975), while in Scania all the hyperites are of one polarity (Bylund, 1981). The K—Ar muscovite ages of 912 and 926 Ma from sheared rocks of the Vaggeryd syenite define the last cooling of the front zone (Patchett, 1978) and a minimum age for the high blocking temperature components in the hyperites.

The predominant 'A' magnetisations in the Bamble sector are linked to ages between 1000 and 1100 Ma and are apparently ca. 100 Ma older than the Rogaland results. They define a coherent swathe which incorporates limited earlier data from these rocks (Poorter, 1975), although the evidence regarding the direction of movement along this path is equivocal. We regard it as representing an earlier part of the overall a.p.w. path for the Sveconorwegian terrain (Fig. 15). The interpretation of the minority 'B' magnetisations is not readily explicable in the context of the known record from the Sveconorwegian terrain (Fig. 15, and Patchett and Bylund, 1977), and discussion is deferred below.

The importance of the palaeomagnetic results from the front-parallel dykes (Patchett and Bylund, 1977) is that they provide a sequence of palaeopoles from events linked to the post-tectonic uplift of the Sveconorwegian terrain which are contemporaneous with magnetisation events inside of the belt; the palaeopoles are linked to Rb—Sr ages between 996 and 869 Ma, but averaging 926 Ma (Patchett, 1978), and plot in a region discrete from all of the Rogaland poles, with the exception of the low blocking temperature 'Y' component, supporting the consensus of the blocking temperature data that they are somewhat younger in age than the remaining poles.

Thus, the collective palaeomagnetic data from the Sveconorwegian belt and its margins define a flattened anticlockwise a.p.w. loop (Fig. 15) confirming the suggestion of Patchett and Bylund (1977) which was based on the limited coverage available at that time. The amplitude of the loop is $\sim 70^\circ$ and it represents in total some 100—200 Ma of a.p.w. movement. The poles from the Brattön and Älgön norites fall with the group of poles from the front-parallel dyke swarm (Fig. 15) although the regional age data suggest that their magnetisations may be up to 150 Ma older. We favour an assignment of these poles to the beginning of the a.p.w. loop with this motion probably also including the post-Älgön dyke. A Sveconorwegian age would fit two of the dyke magnetisations (Listed and Bolshavn primary components) identified by Abrahamsen (1977) on the island of Bornholm; the younger WNW-trending dykes of the Gothenburg area in Sweden genetically linked with metabasites of the Sveconorwegian (Dalslandian) Dal Formation (Larsson, 1956) yield a palaeopole near to the post-Älgön dyke and also the younger limit of this a.p.w. loop in accordance with the inference of Abrahamsen (1974) that they also have a Sveconorwegian remanence.

The distribution of the dolerite poles around the younger limb of the loop is reflected on a local scale by the differences in the magnetisation axes of

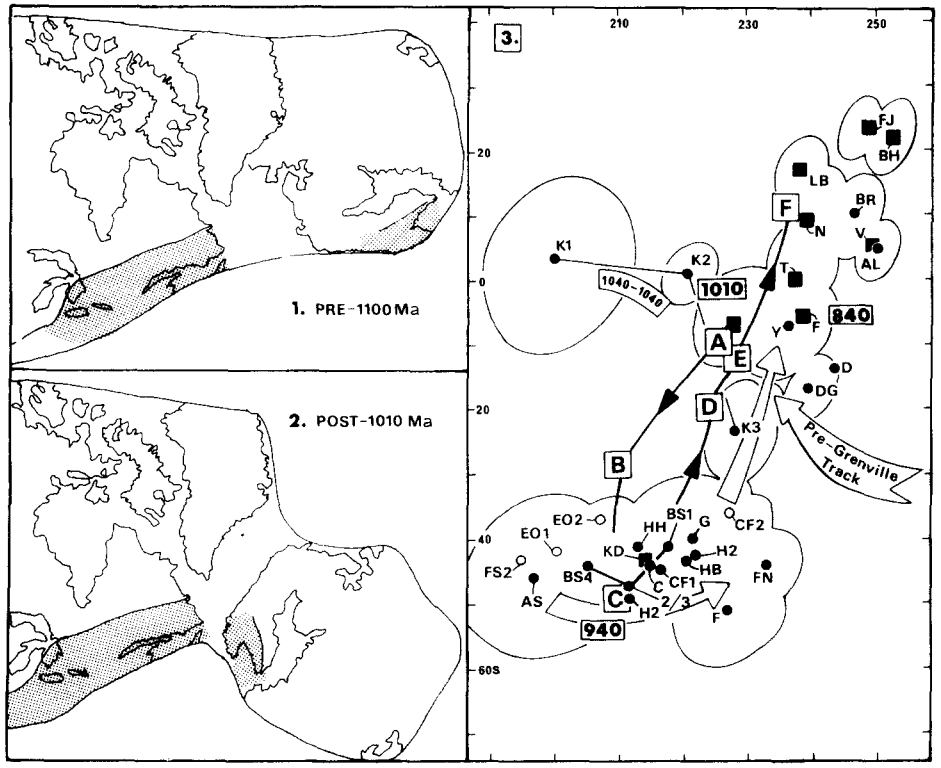


Fig. 15. The reconstruction of the Fennoscandian and Laurentian Shields (1) prior to 1100 Ma and (2) after 1010 Ma, from Patchett and Bylund (1977) and Piper (1982). The ca. 1100 Ma Grenville and Sveconorwegian mobile belts are stippled; the continuation of this belt in Scandinavia is overprinted by the Caledonian orogenic belt and the extension of the Grenville belt in the United States is not entirely clear. The palaeomagnetic data defining reconstruction (2) come predominantly from the cooling-related magnetisations in the Grenville and Sveconorwegian belt and are plotted in (3). The Fennoscandian poles from the Sveconorwegian belt are shown in closed circles (this study) and open circles (previous studies) and palaeopoles from the already-stabilised Svecofennian terrain are shown by closed squares. They are referenced: (i) Sveconorwegian terrain, this study: FS2, farsundite mean; FN, farsundite, northern part; HB, Hidra body; G, Garsaknatt body; AS, Aana-Sira Massif; BS, Bjerkrem—Sogndal lopolith, first phase (1), second and third phase (2/3) and fourth phase (4); C, anorthosite complex, mean all sites, CF1, anorthosite and farsundite complexes, all sites; Y, Rogaland 'Y' component; H2, hyperites (Sweden); K, Bamble—Kragero district, 1, 2 and 3 poles; D, post-Ålgön dyke; BR, Brattön norite; AL, Ålgön norite. (ii) Sveconorwegian terrain, other studies: EO1, Egersund—Ogna body (Hargraves and Fish, 1972); EO2, Egersund—Ogna body (Murthy and Deutsch, 1974); CF2, Rogaland group mean (Poorter, 1972); H2, hyperites (Bylund, 1981); DG, WNW dykes Gothenburg (Abrahamsen, 1974); LB, Listed and Bølshavn dykes (Abrahamsen, 1977). (iii) Results from contemporaneous rocks of stabilised Svecofennian terrain: KD, Karlshamn dyke (936—871 Ma); FJ, Fäjö dyke; BH, Bräkne—Hoby dyke (975—878 Ma); N, Nilstorp dyke (985 Ma); V, Väby dyke; T, Tärnö dyke; HH, Årby dyke (995 Ma); and F, Falun dyke (966—914 Ma). The Grenville palaeomagnetic data are summarised in terms of the mean palaeomagnetic poles A—F computed from the regional chrontour zonation (McWilliams and Dunlop, 1978) and plotted in sequence as open squares; these poles are linked to chrontour ages of A, $\geqslant$ 1000 Ma; B, 1000—950 Ma; C, 950—900 Ma; D, $<$ 900 Ma; E, secondary R and F, secondary N.

the normal and reversely-magnetised hyperite bodies observed by Mulder (1971) and this study. The scale of a.p.w. motion represented by this path suggests that these differences may represent a real time difference in the times of aquisition of NRM rather than the operation of a more complex geomagnetic field model. Later events, notably widespread Palaeozoic dyke intrusion, do not appear to have significantly influenced the Sveconorwegian magnetic and radiometric record in these rocks (Patchett, 1978; Bylund, 1981), but may be responsible for the low blocking temperature component observed in some hyperites of this study and dykes on the island of Bornholm (Abrahamsen, 1977).

Intercontinental correlations

Most discussions of recent years (e.g., Wynne-Edwards and Hasan, 1970; Ueno et al., 1975) have linked the Sveconorwegian belt with the Grenville belt in the Laurentian Shield. Both exhibit a regional K—Ar record in the range 1100—800 Ma, are composed, at least in part, of much older rocks and are separated from earlier-stabilised regions by narrow and near-vertical tectonic boundaries. The magnetisation record in the Grenville terrain has been linked to the cooling history of the terrain (McWilliams and Dunlop, 1978) on the basis of two assumptions, namely that the Grenville poles are of post-metamorphic age and that the cooling history has followed the sequence described by the K—Ar thermochrons (although the actual ages of magnetisation are certainly older than the ages defined by these chrontours). On this basis the palaeomagnetic data points are grouped into 4 zones (A—D) with two secondary components (E and F) to define a flattened loop. This assessment is generally supported by the sequence of high to low blocking temperature components, the remagnetisation of dyke suites, and limited data from elsewhere in the Laurentian Shield, such as the Grand Canyon succession. However, Roy and Blythe-Robertson (1979) discuss several difficulties with this simple cooling model and come to the disappointing conclusion (equally applicable to the Sveconorwegian terrain) that for these uplift-related magnetisations "the mean pole obtained from a palaeomagnetic study of a rock unit is only a rough approximation of the true pole location at the time that the body became magnetised, that is, the pole derived from the sites selected does not necessarily represent the location of the entire rock unit pole as accurately as the statistics indicate". These authors note in particular the scatter of site mean directions from individual units, and the limitations of the available rock exposure and of the vector subtraction approach; it would appear that the rigid interpretation of individual poles is not justified.

In spite of this limitation, there does appear to be a large measure of consistency in the overall trends defined by the Grenville and Sveconorwegian data (Fig. 15). The former come from a province where there is a general south eastwards decrease in ages away from the front zone (Baer, 1976) so that younger poles tend to come from the interior of the belt and define the

later part of the loop defined by McWilliams and Dunlop (1978). This trend is reflected, in part, in the Sveconorwegian belt although the regional effect is complicated here by faulting of, for example, the low grade Telemark terrain (which has not yet been the subject of palaeomagnetic investigation) against the high-grade Bamble terrain; the thermochron approach is not possible here, but the range of regional studies together with the front-parallel dykes leaves no doubt that the overall a.p.w. signature is an anticlockwise a.p.w. loop. Patchett and Bylund (1977) matched the general distribution of Grenville and Sveconorwegian poles by rotating the former anticlockwise by 66.5°, about a pole at 80.5°N 274°E, to derive a reconstruction with the shields in continuity (Fig. 15: 2). The procedure is reinforced by the finding of this study that similar anticlockwise loops are being correlated and no significant modification of Patchett and Bylund's reconstruction appears to be merited. The Bamble sector 'A' magnetisations at ca. 1040—1010 Ma correlate with the Grenville 'A' Zone (ca. 1000 Ma) magnetisations; the 940—910 Ma Rogaland swathe correlates with the Grenville 'C' Zone magnetisations (950—900 Ma) and suggests that the 980 Ma estimate for the Haliburton 'B' magnetisation (Berger et al., 1979) may be marginally too old. The Grenville 'D' (< 900 Ma) and 'E' zones correlate with the Rogaland 'Y' magnetisation and front-parallel dyke palaeopoles in accordance with the 820 Ma estimate on the Haliburton 'C' component (Berger et al., 1979). There is still, however, a real discrepancy in the age estimates of this latter part of the a.p.w. path because the front-parallel dykes have ages mostly in excess of 900 Ma which are likely to be reliable indicators of the magnetisation age (Patchett and Bylund, 1977; Patchett, 1978), and the Brattön and Älgön bodies are almost certainly older than this; it would appear that either the younger ages are underestimates or that execution of the loop was complete by 900 Ma and was followed by a period of little or no a.p.w. movement.

The palaeomagnetic reconstruction of the Fennoscandian and Laurentian Shields during the post-tectonic uplift and cooling of the Grenville and Sveconorwegian terrains has the Grenville Front and the Schistosity Zones disposed approximately at right-angles to one another. These front zones are probably related to the post-tectonic uplift and are genetically linked to fracturing and dyke intrusion in the Fennoscandian Shield (Patchett and Bylund, 1977); the Grenville Front also involves limited sinistral strike-slip movement (McWilliams and Dunlop, 1978). It is highly unlikely that these two lineaments were initiated at this high angle to one another and indeed, the pre-Grenville/Sveconorwegian a.p.w. records from the two shields do not accord with one another on this reconstruction (Patchett et al., 1978). Instead they accord with a primitive reconstruction which has the two front zones in alignment (Patchett et al., 1978; Piper, 1982), an accordance which suggests that this lineament originated before the 1050—850 Ma configuration of Fig. 15 was attained. The youngest result to accord with this primitive reconstruction are the post-Jotnian dolerites dated ca. 1270—1190 Ma and

there is as yet no equivalent in the Fennoscandian Shield to the ca. 1100—1040 Ma Keweenawan record; until this is forthcoming it will not be possible to confirm that these fronts originated in alignment. It may, however, be significant that the 'B' poles from the 1090—1010 Ma intrusions of the Bamble sector, which correlate with no part of the Laurentian record on the post-1010 Ma reconstruction, fall on the a.p.w. path at ca. 1100 Ma on the primitive reconstruction [rotated positions are 186°E 60°N and 232°E, 39°N (see Piper, 1982, figs. 5 and 6)]. Thus, it is possible that the magnetite-held record here is a relict which sees through the break up and rotation of these shields prior to uplift and cooling of the mobile terrains; this suggestion is no more than one possible solution because it requires that these magnetite-held remanences pre-date higher blocking temperature, predominantly hematite-held, 'A' remanences. This area merits further study before this problem can be resolved.

The geological implications of the break up and relative movement of the two shields at ca. 1100 Ma have been discussed in some detail elsewhere (Patchett et al., 1978; Piper, 1983). They include protracted and episodic alkaline magmatism for some 200 Ma prior to break up, widespread rifting and intrusion of the Mackenzie and Jotnian, shield-wide, dolerite episodes and the formation of grabens (notably the Keweenawan) related to strike-slip movements along the front zones.

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